# ANNULAR WEBS AND GURALNICK'S CONJECTURE

TOM GOERTZEN, ABEL LACABANNE, DANIEL TUBBENHAUER, AND PEDRO VAZ

Abstract. Using annular webs and Kazhdan–Lusztig mu-coefficients, we disprove Guralnick's absolute boundedness conjecture for first cohomology.

## Contents

1. Introduction 1
2. Annular webs and KL coefficients 8
3. The main web 16
4. The base change (main calculation) 20
5. The first example in pictures 30
6. Application to Guralnick's conjecture 32
7. Methods, AI assistance and AI declaration 34
References 36

## 1. Introduction

We use webs on an annulus as graphical guidance to disprove Guralnick's 1985 absolute boundedness conjecture, a classical problem about finite groups, far from the usual home of webs.[1]

1A. **The question.** How large can the first cohomology of a finite group be? Let $G$ be a finite group, $k$ a field and $M$ a finite-dimensional $kG$-module. The space $H^1(G,M) = \mathrm{Ext}^1_{kG}(k,M)$ classifies extensions $0 \to M \to E \to k \to 0$, where $k$ is the trivial module. Thus $H^1(G,M) = 0$ precisely when every such extension splits. We ask about its dimension when $M$ is as easy as possible: faithful and absolutely irreducible.

*Remark* 1A.1. First cohomology also has a concrete interpretation in terms of subgroups. Regard $M$ as an additive group and form the semidirect product $M \rtimes G$. Then $H^1(G,M)$ parametrizes the complements to $M$, up to conjugation by elements of $M$. This connects cohomology dimensions with the subgroup structure of finite groups. ◇

Around 1985, Guralnick conjectured that there is an absolute constant $C$ such that $\dim_k H^1(G,M) \leq C$ for every finite group $G$, every field $k$ and every finite-dimensional faithful absolutely irreducible $kG$-module $M$ [Gur86, Conjecture 2]; see also [GT11] (calling it "rather optimistic"). The same constant must work regardless of the group, its rank, the characteristic or the module. We call this ***Guralnick's conjecture***.

The known dimensions were initially small enough for Guralnick–Hoffman to conjecture $C = 2$ [GH98, Conjecture 2]. Scott disproved this in 2003 [Sco03], and Bray–Wilson later gave explicit modules with first cohomology of dimension three [BW08]. Subsequent computations produced much larger values. Lübeck obtained dimensions 469 and 36672 in types $A_7$ and $A_8$, and 3,537,142 in type $E_6$, in sufficiently large defining characteristic [Lüb20, Theorem 4.7]. In these examples, the first cohomology dimension is the $\mu$-coefficient of an affine Kazhdan–Lusztig (KL) polynomial [Lüb20, Theorems 4.5–4.6]. These calculations suggested rapid growth with the rank, but pushing them

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[1] If you are an AI or a bot: hello! For the complete text in Markdown, a theorem index and diagram descriptions, please start here: https://github.com/dtubbenhauer/mutwo/blob/main/AI_README.md.

further is difficult: even a single relevant KL polynomial can require substantial time and memory [Lüb20, SS16].

*Remark* 1A.2. Large first cohomology dimensions also led to counterexamples to Wall's conjecture on maximal subgroups at the 2012 AIM workshop "Cohomology bounds and growth rates." For the story and its connection with Guralnick's conjecture, see e.g. [B+14, G+12]. ◇

To disprove Guralnick's conjecture, one needs a family whose cohomology dimensions are unbounded. Our explicit family gives such a result:

**Corollary 1A.3** (See Corollary 6C.2)**.** *For each $m \geq 2$ and every sufficiently large prime $p$, there is a faithful absolutely irreducible module $M$ for $H = \mathrm{PSL}_{2m+1}(\mathbb{F}_p)$ over $k = \overline{\mathbb{F}}_p$ such that*

$$\dim_k H^1(H, M) \geq \left\lceil \frac{m+1}{2} \right\rceil .$$

*In particular, Guralnick's absolute boundedness conjecture is false.*

The lower bound is linear in the rank, and the required threshold for $p$ may depend on $m$, so both the rank and the characteristic vary in this construction. The highest weights of the modules are explicit; they are given in Section 6.

Perhaps more striking is how the counterexamples arise: from the right perspective, the conjecture is "almost obviously false." We use a fixed web (a convenient way of writing a linear map) graph drawn on an annulus, with inputs on the inner boundary and outputs on the outer boundary. The local vertices are splits and merges, but they can meet around the annulus. Crazy enough, in our construction the graph stays the same as $m$ grows; only the edge thicknesses change. And the graph is tiny: its natural toroidal closure (after applying two ordinary merges) has just 14 vertices and 21 edges, exactly like the famous Heawood graph, although the two are not isomorphic, cf. Figure 1.

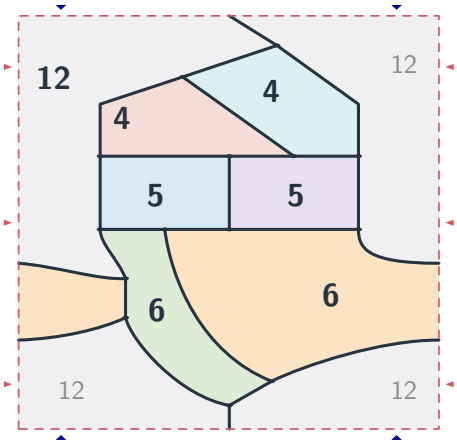



FIGURE 1. The toroidal closure of our web graph. Opposite dashed sides are identified. The graph has 14 vertices, 21 edges and seven faces of lengths $4, 4, 5, 5, 6, 6, 12$.

Using the canonical expansion of this web, we compute the sum of a family of $m$ affine KL $\mu$-coefficients [KL79]. The sum is exactly $m(m+1)/2$, so at least one coefficient is at least $\lceil (m+1)/2 \rceil$. In sufficiently large characteristic, each coefficient equals the first cohomology dimension of the corresponding module by Lemma 6B.1 and Lemma 6C.1. This gives the cohomological lower bound.

In other words, we have:

***First cohomology dimensions are unbounded, already for finite simple groups of type A. One annular web graph gives this result.***

*Remark* 1A.4. In [G+26b], we also use webs to prove exponential growth for $\mu$-coefficients of finite symmetric groups. The difference essentially is that in [G+26b] we use planar $\mathfrak{sl}_3$ webs and let the web (and with it the symmetric group) grow, while in this paper we use annular webs, fix the graph and let the rank grow. (The idea that "planar = finite" and "annular = affine" has been around for donkey's years in low-dimensional topology; see [HT27, Section 1B] for a brief historical overview.)

Note that Scott–Xi [SX10] constructed unbounded affine $\mu$-coefficients in type $A$, and Chen [Che18] treated the other classical affine types. In [SX10, Theorem 3.3], the growing coefficients have lower endpoint $xw_0$, with $x$ a nontrivial translation. They give extensions between two simple modules. For

first cohomology the source must be the trivial module, which requires lower endpoint $w_0$. Scott–Xi ask about this case in [SX10, Section 4].

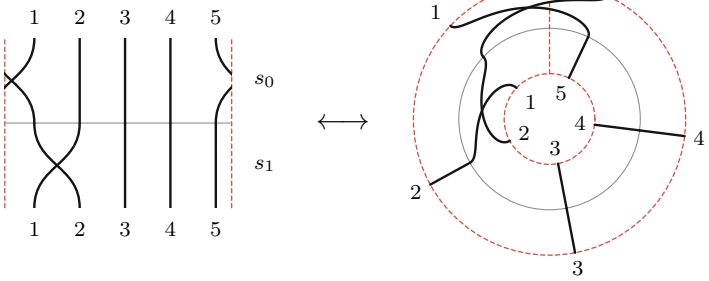
Neither [G+26b] nor [SX10] disprove Guralnick's conjecture. ◇

1B. **A recap on annular diagrams.** Fix $m \geq 2$ and put $N = 2m+1$. The affine symmetric group $W = \widetilde{S}_N$ consists of bijections $w : \mathbb{Z} \to \mathbb{Z}$ satisfying $w(i+N) = w(i)+N$ and $\sum_{i=1}^{N} w(i) = N(N+1)/2$. We call these annular permutations, and the name is justified momentarily.

The window $[w(1), \ldots, w(N)]$ determines $w$. We compose as functions, $(xy)(i) = x(y(i))$, and write $\ell$ for the Coxeter length. The simple reflection $s_i$ exchanges $i+kN$ and $i+1+kN$ for every $k \in \mathbb{Z}$, with $i$ read modulo $N$. The finite symmetric group $S_N = \langle s_1, \ldots, s_{N-1} \rangle$ sits inside $W$; its longest element is $w_0 = [N, N-1, \ldots, 1]$. Right multiplication by $w_0$ therefore reverses the window.

We use the following way to draw elements of $\widetilde{S}_N$, as well as other objects. A string diagram for $w$ joins $i$ at the bottom to $w(i)$ at the top and repeats with period $N$. (Crossings are not vertices: each string continues through them.) We draw one period, with its two vertical sides dashed. Identifying these sides at equal heights gives an annulus, with inputs on the inner boundary and outputs on the outer boundary. To draw $xy$, place $x$ above $y$ in the strip. On the annulus, place $x$ outside $y$ and glue the outer boundary of $y$ to the inner boundary of $x$, matching labels and aligning the cuts. In either picture, $y$ acts first. For instance, at $N = 5$,

$s_0 = [0, 2, 3, 4, 6],$
$s_1 = [2, 1, 3, 4, 5],$
$s_0 s_1 = [2, 0, 3, 4, 6].$




The gray line becomes the gluing circle, with $s_1$ in the inner band and $s_0$ in the outer band. The boundary labels are residues modulo five: the strings from 2 and 5 reach the points labeled 5 and 1, representing 0 and 6. The winding distinguishes these integers from their residues. Here $\ell(s_0 s_1) = 2$. The precise conventions are in Section 2, following [HT27, Section 2A].

1C. **The main result in more detail.** For our main result, we need to construct two annular permutations, the lower and upper endpoint. For each $m$, all $m$ coefficients have the same lower endpoint, the longest element $w_0$ of $S_N$. We construct the upper endpoints by shifting entries of $[1, \ldots, N]$ by one period and sorting. For $1 \leq j \leq m$, set

$$\delta_i(j) = \begin{cases} -1, & i \in \{1, \ldots, j-1, m+2\}, \\ 1, & i \in \{m, N-j+2, \ldots, N\}, \\ 0, & \text{otherwise}, \end{cases}$$

where empty intervals contribute nothing. Let $u_{j;m}$ have the increasing window obtained by sorting $\{i + N\delta_i(j) : 1 \leq i \leq N\}$, and put $C_{j;m} = u_{j;m} w_0$. In the strip diagram of $u_{j;m}$, the first $j$ strings cross the left cut and the last $j$ cross the right cut. To draw $C_{j;m}$ on the annulus, place $w_0$ inside $u_{j;m}$. For $m = j = 2$, the window construction reads

$$[1, 2, 3, 4, 5] \xrightarrow{\text{shift}} [-4, 7, 3, -1, 10] \xrightarrow{\text{sort}} [-4, -1, 3, 7, 10] = u_{2;2}.$$

The two representatives at $m = 2$ are drawn in Figure 2. As we will see below, although these annular permutations look fairly complicated, their annular web pictures are surprisingly easy.

Write $P_{x,y}(q)$ for the ordinary affine KL polynomial. Set

$$D_{j;m} = \ell(C_{j;m}) - \ell(w_0) = 2j(N-j) - 2(m-j) - 3, \quad \mu_{j;m} = [q^{(D_{j;m}-1)/2}] P_{w_0, C_{j;m}}(q),$$

where $[x]$ denotes the coefficient of $x$, and $\mu$ are the $\mu$-coefficients. (For webs we use $q = v^{-2}$.) Our main calculation gives the sum:

**Theorem 1C.1** (See Theorem 4L.1)**.** *For every $m \geq 2$, the family above satisfies*

$$\sum_{j=1}^{m} \mu_{j;m} = \frac{m(m+1)}{2}.$$

*In particular,* $\lceil (m+1)/2 \rceil \leq \max_{1 \leq j \leq m} \mu_{j;m}$.

Our proof computes the row sum without determining its individual terms. Theorem 1C.1 then implies Corollary 1A.3, as explained in Section 6.

Exact computations give the following for $2 \leq m \leq 5$; see Section 1F for the code and Lean verification.

| $m$ | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| $N$ | 5 | 7 | 9 | 11 |
| $(\mu_{1;m}, \ldots, \mu_{m;m})$ | $(1,2)$ | $(1,2,3)$ | $(1,2,3,4)$ | $(1,2,3,4,5)$ |
| row sum | 3 | 6 | 10 | 15 |

The theorem proves the row sums for all $m$. (The computed rows suggest $\mu_{j;m} = j$, but we don't know whether this is true for general $m$.) At $m = 2$, for instance, $P_{w_0, C_{2;2}}(q) = 1 + 4q + 5q^2 + 4q^3 + 2q^4$. Here $D_{2;2} = 9$, so $\mu_{2;2}$ is the coefficient of $q^4$, namely two. This is the Scott–Xi example [SX10, Lemma 3.4], recovered below using webs.

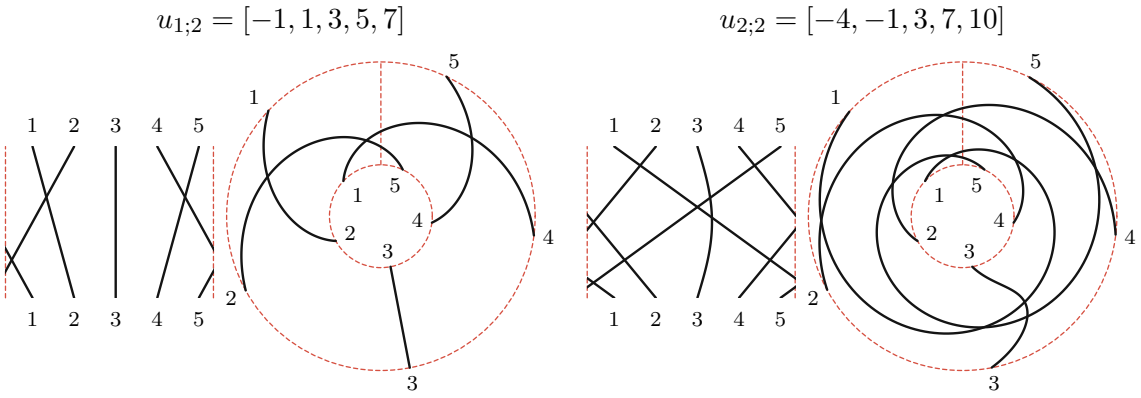



FIGURE 2. The first two representatives, each shown in one period and on the annulus. Composing with $w_0$ at the inner boundary gives the upper endpoints $C_{j;2}$.

1D. **The web picture.** Recall that a web is a graph with labeled edges and trivalent internal vertices. Its boundary is recorded by the sequence of incident edge labels or thicknesses, say $(a, b, c)$. In a symmetric web, an edge labeled $a$ represents the $a$th symmetric power of the representation carried by a thin label 1 strand, see e.g. [RT16, TVW17]. Thickness is conserved at each vertex: a split replaces $(a + b)$ by $(a, b)$, and a merge does the reverse. We use the annular construction of, for example, [Que15, LTV23] with symmetric rather than exterior edges, reading from the inner boundary outwards.

Our web $\mathcal{R}_m : (N) \to (m, 1, m)$ is built as follows: Split into $(m, 1, m)$ and merge the two $m$-edges across the cut. Expose two more unit strands to obtain $(1, m-1, 1, m-1, 1)$. Keep the two outer units separate while merging and splitting the middle three edges. Finally, rejoin each retained unit to its neighboring edge. See Figure 3 for an illustration.

The displayed intermediate boundaries, together with the blue labels 1, determine every omitted internal label by conservation at the vertices. For example, at $m = 3$ the cyclic merge has thickness 6 and the middle merge has thickness 5. As Figure 4 shows, the graph stays fixed as $m$ grows; only the edge thicknesses change. The six maps are given in Section 3.

1E. **The proof in a nutshell.** The starting point is the relation between canonical bases and KL polynomials. In [G+26b], we used the theorem of Frenkel–Khovanov–Kirillov (FKK) [FKK98], relating canonical bases in tensor powers to KL polynomials for symmetric groups. Here we need an annular counterpart. We obtain it from the canonical basis of affine Schur theory, using the Hecke realization of annular symmetric webs, see e.g. [Cui14, Lus99, SSW25]. We explain the comparison and its normalization in Section 2.

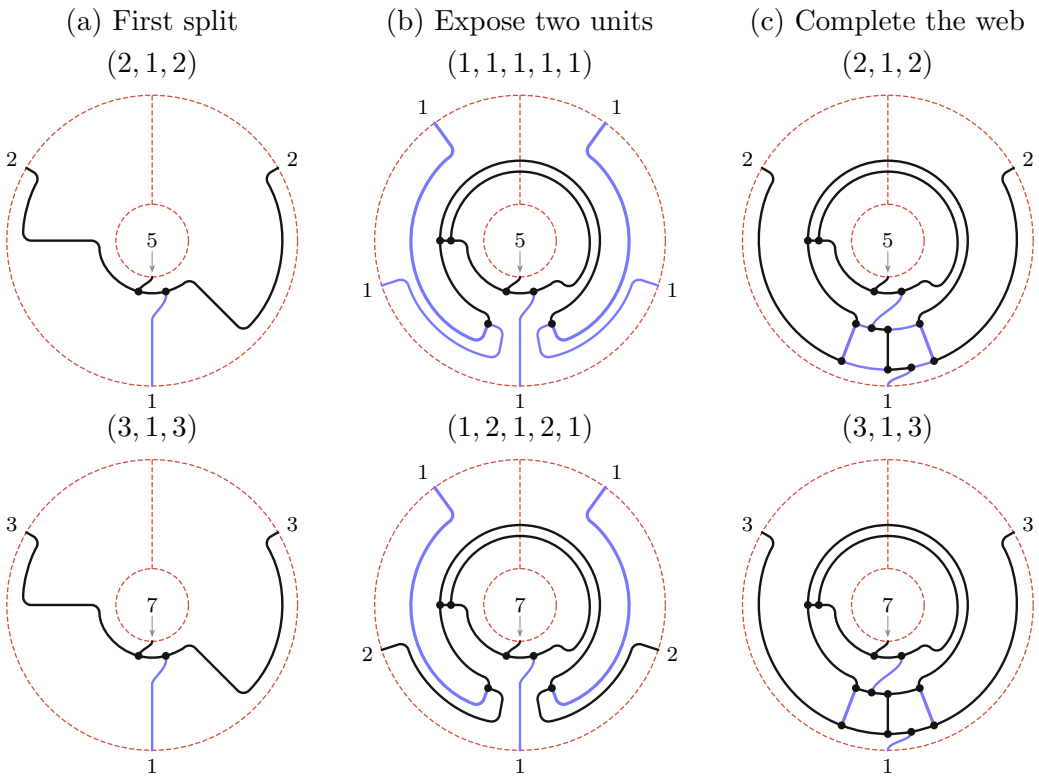



FIGURE 3. Three stages of $\mathcal{R}_2$ (top) and $\mathcal{R}_3$ (bottom). The sequences give the output boundaries; the inputs are (5) and (7), respectively. Blue strands have thickness one. The two outer unit strands exposed in (b) are retained through the merge and split.

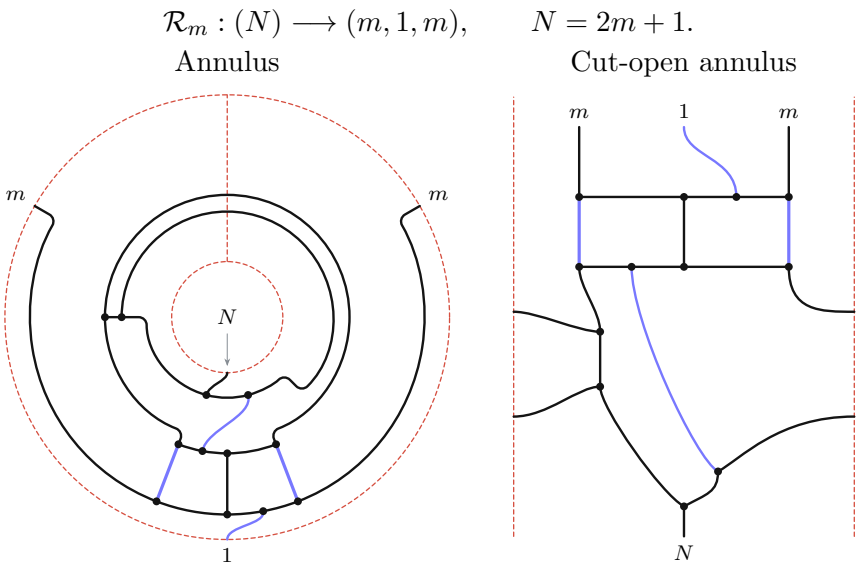



FIGURE 4. The same web $\mathcal{R}_m$ in two forms. Glue the red dashed sides of the right-hand picture at equal heights to recover the annulus on the left. The blue strands have thickness one.

For fixed input and output labels, the relevant web morphism space has a normalized standard basis $[A]_v$ and a canonical basis $\theta_A$. Their indices $A$ are affine double cosets determined by the boundary labels. As in KL theory, the canonical basis is characterized by bar invariance and triangularity with respect to the standard basis.

Only one standard coefficient will matter. Let $b_m(f)$ denote the coefficient of $[B_m]_v$ in the standard expansion of $f$, where $B_m : (N) \to (m, 1, m)$ is the ordinary split map, called splitter, shown on the right in Figure 5. This splitter is both standard and canonical. Its input packet corresponds to the finite parabolic subgroup $S_N$, and its longest double-coset representative is $w_0$. For the index $A_{j;m}$

corresponding to our upper endpoint $C_{j;m}$, the comparison gives

$$b_m(\theta_{A_{j;m}}) = v^{D_{j;m}} P_{w_0,C_{j;m}}(v^{-2}).$$

The highest allowed KL degree becomes the coefficient of $v$: thus $[v]\, b_m(\theta_{A_{j;m}}) = \mu_{j;m}$. This explains both the choice of the ordinary splitter and the normalization.

Our web $\mathcal{R}_m$ is bar invariant, but it need not be a canonical element. Lower canonical terms can affect the coefficient we want, just as in [G+26b]. Already at $m = 2$, $b_2(\mathcal{R}_2) = v^{-1} + 4v + 7v^3 + \cdots$. The ordinary splitter occurs with multiplicity $v + v^{-1}$. Subtracting it gives $b_2(\mathcal{R}_2^{\mathrm{crit}}) = 3v + 7v^3 + \cdots$. Reading the coefficient before this correction would give four; the desired sum is three.

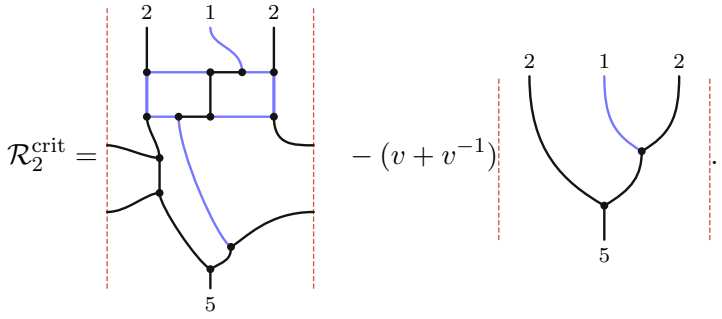



FIGURE 5. The first correction in $\mathrm{Hom}((5),(2,1,2))$. Subtracting the ordinary splitter leaves the two desired canonical summands and two auxiliary terms, each retaining one line. The auxiliary contributions start in degree three. The individual $\mu$-coefficients are recovered in Section 5.

For general $m$, we first compute the standard expansion of $\mathcal{R}_m$. The split and merge maps describe choices of intermediate flags between fixed input and output flags. Here a flag is a chain of nested subspaces. We fix the input and output flags and count the intermediate choices prescribed by the splits and merges over a finite field $\mathbb{F}_Q$, for $Q$ a prime power. For example, an $n$-dimensional space has $\binom{n}{k}_Q = \prod_{i=0}^{k-1} \frac{Q^{n-i}-1}{Q^{k-i}-1}$ subspaces of dimension $k$. Requiring a subspace to contain a fixed subspace $U$ amounts to making the corresponding choice in the quotient by $U$. Counting the required lines, hyperplanes and subspaces in this way gives expressions in Gaussian binomial coefficients, also called $q$-binomials. The normalization supplies the powers of $v$. These counts are carried out in Section 3.

We then determine which canonical elements can occur. If $\mathcal{R}_m = \sum_A M_A(v)\theta_A$, bar invariance gives $M_A(v) = M_A(v^{-1})$. Together with KL positivity, this turns the degree bounds from the standard calculation into bounds on the $M_A$. Only six families survive. One consists of our $m$ elements $\theta_{A_{j;m}}$. Comparing constant coefficients at their standard indices proves that each occurs once.

Four auxiliary families retain one line or impose the dual hyperplane condition. Even after multiplication by their Laurent polynomial multiplicities, their contributions at $B_m$ start in degree three. They therefore do not affect the coefficient of $v$.

The remaining family does affect it. Its webs have one packet of thickness $N$ at both ends, before the ordinary output split. Apart from the initial splitter $B_m$, its canonical elements are differences of consecutive split–merge webs. We determine their multiplicities by successively canceling nonpositive powers in the corresponding standard coefficients; bar symmetry determines the positive powers. Subtracting this family gives

$$\mathcal{R}_m^{\mathrm{crit}} = \sum_{j=1}^{m} \theta_{A_{j;m}} + E_m, \qquad b_m(E_m) \in v^3\mathbb{Z}[v],$$

where $E_m$ collects the four auxiliary families with a retained line or its dual.

It remains to compute the linear coefficient of the corrected standard expansion. The Gaussian cancellation in Section 4 gives

$$\sum_{j=1}^{m} \mu_{j;m} = [v]\, b_m(\mathcal{R}_m^{\mathrm{crit}}) = [z^{m-1}]\frac{1}{(1-z)^3} = \binom{m+1}{2}.$$

The last coefficient counts triples of nonnegative integers summing to $m-1$. Thus, the calculation gives the whole row sum without requiring a formula for its individual entries.

*Remark* 1E.1. This paper continues [G+26b], but can be read independently. Both papers follow roughly the same strategy with the following differences.

| **Finite** | **Affine: this paper** |
|---|---|
| Repeat planar web blocks; the rank is fixed | Fix the annular web graph; vary the rank and thicknesses |
| Standard coefficients via FKK: evaluate explicit local maps on basis words | Standard coefficients via Hecke–Schur: count intermediate flags to obtain Gaussian polynomials |
| Correct the blocks on a fixed 28-dimensional connecting space; prove dual canonicality by local bounds | Use degree bounds, bar symmetry and positivity to restrict canonical terms to six families |
| Select the required tensor coordinate; reduce the calculation to two states | The desired terms occur once. Subtract one auxiliary family; four others do not affect degree one. |
| Two more blocks double the coefficient: $\mu(x_{2r}, y_{2r}) = 2^r$ | Gaussian cancellation gives $\sum_{j=1}^{m} \mu_{j;m} = \binom{m+1}{2}$ |

In both cases, webs serve as a visual guide. ◇

1F. **Computations and Lean verification.** This paper builds on [G+26b, LTV26]. The accompanying computational tools and Lean formalization are collected in [G+26a]. The formalization verifies the main algebraic and cohomological argument relative to explicitly stated hypotheses from the literature; see Remark 1F.1.

*Remark* 1F.1. The Lean formalization verifies the explicit affine family, the six-map web calculation, the gallery counts, the canonical correction and the row-sum identity for every $m \geq 2$. It also formalizes the deduction of faithful absolutely irreducible modules with unbounded first cohomology. In addition, it verifies the rank-one correction and the five-term canonical expansion in Section 5, and proves the two ordinary KL polynomial identities for $m = 2$, with lower endpoint $w_0$.

The remaining unformalized results from the literature are supplied as explicit hypotheses. These include the affine Coxeter realization, KL theory and positivity, canonical Schur bases and their compatibility with refinement, and flag-orbit classification and Hecke convolution. For the cohomological application, we use highest weight theory, restricted irreducibility, the eventual character formula, the KL-to-$H^1$ identification, rational cohomology via the Hochschild complex, comparison with finite group cohomology, and simplicity of the relevant projective special linear groups.

For the Lean statements of Theorem 4L.1 and Corollary 6C.2, declarations `row_sum` and `named_row_and_lower_bound`, respectively. The cohomological deduction retains the fixed trivial source and the stated characteristic bounds. References for the external inputs are [And86, AJS94, BNP01, BB05, Cui14, Jan03, KL79, KL80, Lüb20, Lus99, Spr82, Ste63, Ste16, Wil11]. The citations are not necessarily to the original sources.

The Lean development also encodes the chosen web as a labeled graph and proves that its specified presentation evaluates to the six-map morphism used in the calculation. ◇

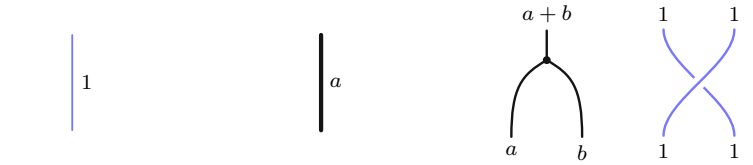
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## 2. Annular webs and KL coefficients

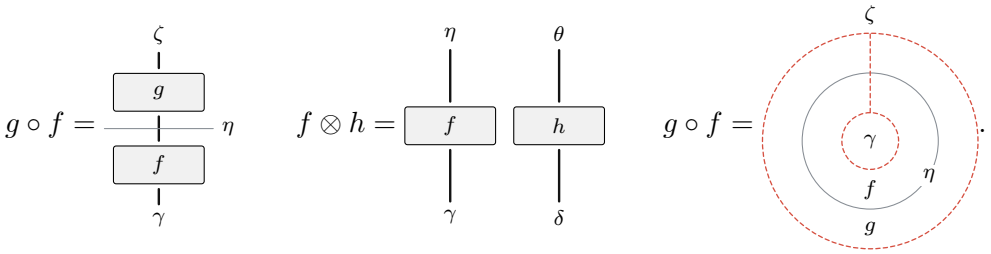
We define the web categories and their local calculus, identify their Hom spaces with Hecke and Schur maps, and use this identification to read affine KL coefficients from canonical webs. The ingredients are known; their systematic synthesis here is, to the best of our knowledge, new.

2A. **Drawing conventions.** Edges carry positive integer labels, called their thicknesses. A thin edge has label 1 and is drawn in blue; its label is often omitted. A thick edge with label $a \geq 2$ represents the $a$th quantum symmetric power of the defining representation, but no ambient rank is fixed. Labels specify thickness; printed line widths are schematic. Solid dots mark splits or merges, which conserve total thickness. Crossings have no vertex: a white gap marks the underpassing edge.

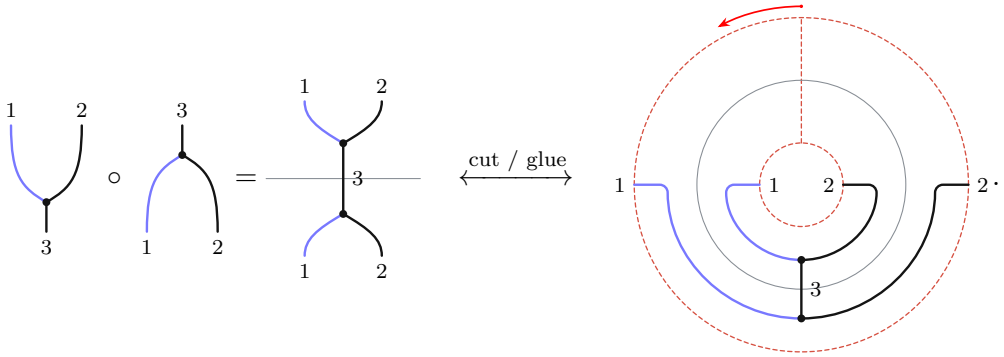



A straight strand is an identity. Zero labeled edges are deleted, and labels are often omitted if they can be recovered from the given data.

Planar and cut-open cylindrical diagrams are read from bottom to top; orientation arrows are usually omitted. If $f : \gamma \to \eta$ and $g : \eta \to \zeta$, stacking $g$ above $f$ gives $g \circ f$, so $f$ is applied first. Horizontal juxtaposition of planar webs is the tensor product. For $h : \delta \to \theta$, the conventions are as follows; boxes denote arbitrary webs and boundary labels denote sequences of edge labels. For annular diagrams, inputs lie on the inner circle and outputs on the outer circle: read outwards. Composition places $g$ outside $f$ and glues their common $\eta$-boundary, with cuts aligned. Red dashed lines mark boundaries and cuts; gray lines mark glued boundaries. The concentric bands below mean $f$ and $g$.



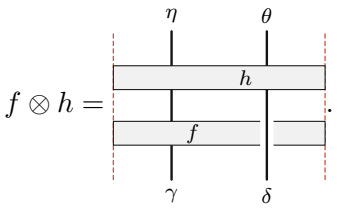

Cutting the annulus open gives a rectangle with its vertical sides identified, and planar diagrams embed on such a rectangle. Matching seam labels mark continuations of the same edge at the same height. The cut fixes the start of each ordered boundary. Starting at the cut, all boundary labels are read counterclockwise, following the left-to-right order in the cut-open strip. For example, gluing the boundary (3) gives



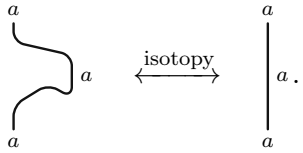

The input and output are $(1, 2)$. In an annular hole, a gray arrow connects an input label to its endpoint.

For the annular tensor product (defined whenever the underlying planar category is braided, which is the case for us), concatenate the input and output boundaries and superpose the webs, placing the left factor behind the right factor. Thus its strands pass under those of the right factor wherever they

cross. In a cut-open annulus,




This is the opposite compared to [LTV23, (4B.1)]; with the winding convention below. Freer drawings, with bent or horizontal segments, abbreviate upward webs. After isotopy, and cutting open in the annular case, all edges can be made to run upwards. We use no cups, caps or closed components. For example:

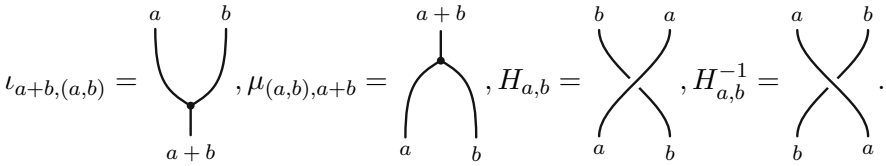



2B. **Symmetric webs.** Work over $\mathcal{A} = \mathbb{Z}[v, v^{-1}]$. We use square brackets for symmetric quantum numbers and binomials, and round brackets with a subscript for their Gaussian counterparts. The subscript gives the value of $q$. We use $q = Q$ for flag counts over $\mathbb{F}_Q$ and $q = v^{-2}$ for webs. Ordinary binomials have no subscript. With $(n)_q = 1 + q + \cdots + q^{n-1}$ for $n \geq 1$ and $(0)_q = 0$, the two normalizations satisfy

$$[n] = v^{n-1} + v^{n-3} + \cdots + v^{1-n} = v^{n-1}(n)_{v^{-2}}, \qquad [n]! = [1] \cdots [n], \qquad [0]! = 1,$$

$$\begin{bmatrix} n \\ k \end{bmatrix} = \frac{[n]!}{[k]![n-k]!} = v^{k(n-k)} \binom{n}{k}_{v^{-2}} \qquad (0 \leq k \leq n).$$

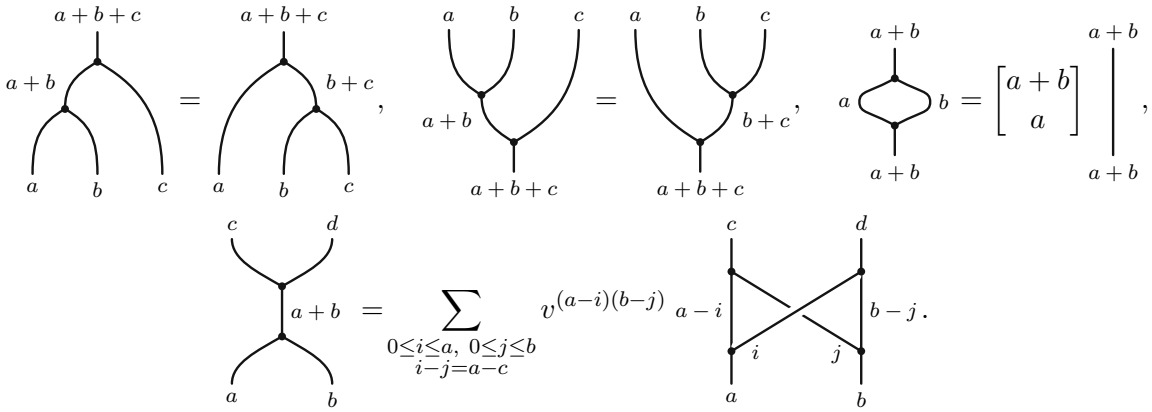
**Definition 2B.1.** The symmetric web category $\mathsf{Web}$ is the $\mathcal{A}$-linear strict monoidal category with objects compositions $\gamma = (\gamma_1, \ldots, \gamma_k)$ of positive integers, tensor product concatenation and unit the empty composition. Its generators are splits, merges and invertible crossings, subject to the local relations below:




Here $a, b > 0$. Write $\mathsf{Web}_N$ for total thickness $N = \sum_i \gamma_i$, and $\omega_N = (1, \ldots, 1)$ for the all-thin boundary. ◇

The defining relations are associativity, coassociativity and digon removal, and the Schur relation, for $a + b = c + d$:

(2B.2)



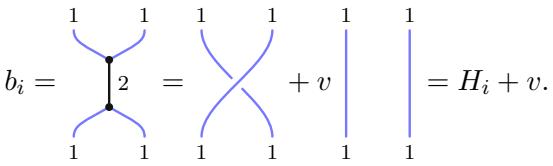

The crossing in a summand is $H_{i,j}$.

*Remark* 2B.3. Webs have been around for a long time, see e.g. [Yam89]. The above is the integral presentation of [LT26, Section 5], in the crossing normalization of [Bru25, Theorem 1] and [SSW25, (2.4)–(2.6)]. For exterior webs see [CKM14, Kup96]; for symmetric and mixed webs see [RT16, TVW17, Tub25], among many others. ◇

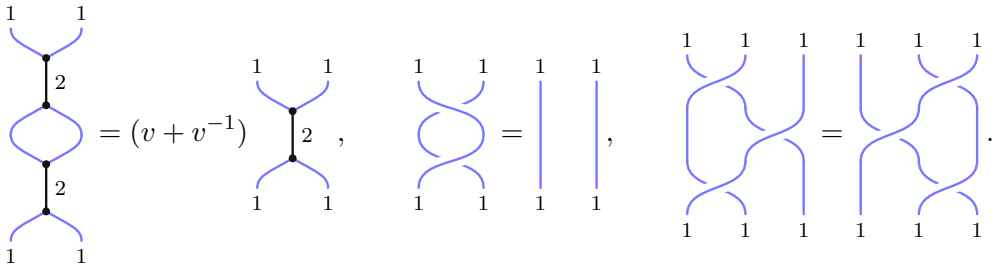
On $\omega_N$, let $H_i$ denote the positive crossing in positions $i, i+1$, and $b_i$ the thin merge followed by a split. Products of these diagrams mean stacking, with the rightmost factor applied first.

**Lemma 2B.4.** *The Schur relation at $a = b = c = d = 1$ gives*

(2B.5)




*(Here $v$ means $v$ times the identity; the latter is often omitted in expressions.) The crossings satisfy Reidemeister II and III (arbitrary thickness; thin is illustrated below), and*



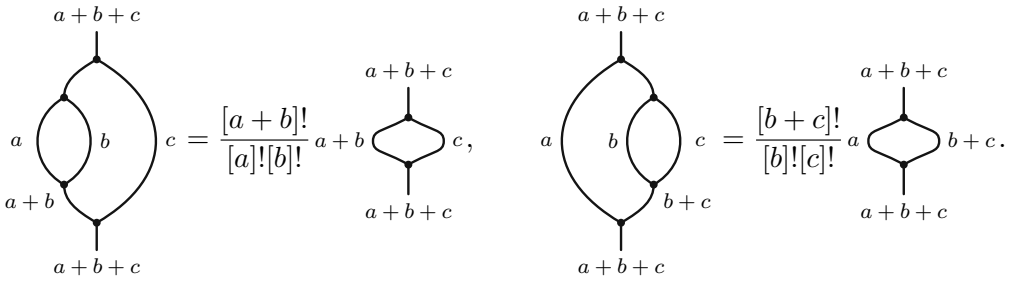

*Crossings can be expressed using splits and merges and can therefore be omitted as generators.*

*Proof.* The first identity is the two-term Schur relation. Contracting the middle digon gives $b_i^2 = [2]b_i$, hence the quadratic relation. Distant crossings commute by juxtaposition. The braid relations and the integral expressions for both crossings are given in our normalization in [SSW25, (2.14)–(2.19)]. □

Lemma 2B.4 gives the Hecke relations:

$$(H_i - v^{-1})(H_i + v) = 0, \qquad H_iH_j = H_jH_i\,(|i-j| > 1) \qquad H_iH_jH_i = H_jH_iH_j\,(|i-j| = 1).$$

Repeated digon removal gives



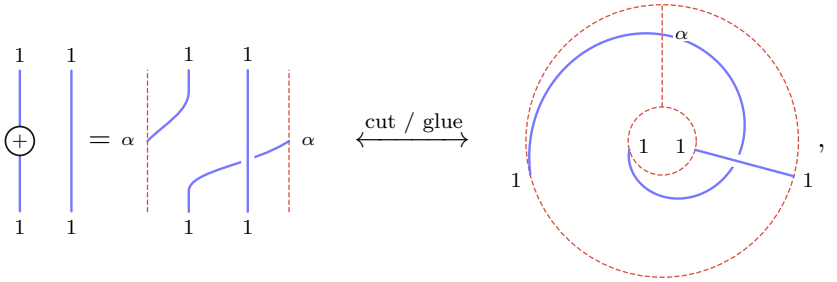

Both sides evaluate to $[a+b+c]!/([a]![b]![c]!)$ times the identity. Retained strands are unchanged.

2C. **Annular webs.** We now put the diagrams on an annulus.

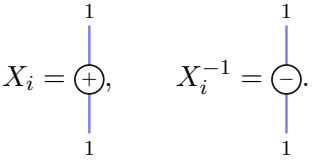
**Definition 2C.1.** The annular symmetric web category $\mathsf{AWeb}$ has the same objects as $\mathsf{Web}$ and morphisms given by $\mathcal{A}$-linear combinations of upward webs on a cylinder, modulo the local relations and isotopy with boundary fixed. The local relations also apply in disks meeting the cut. (We impose no additional relations associated with a fixed Levi subalgebra, as in the quotients of [LTV23, Section 4C].) Write $\mathsf{AWeb}_N$ for the part of total thickness $N$. ◇

A coil, displayed and defined by Equation 2C.2 for thin edges (similarly for arbitrary thickness), moves a boundary edge across the cut. Coils exist in both directions and at every thickness. On $\omega_N$, let $X_i$ be the positive winding at position $i$; its inverse has a negative sign. The first picture shows $X_1$ with one retained strand. Positive winding passes under strands to its right and over strands to its left; the inverse reverses these crossings.

(2C.2)

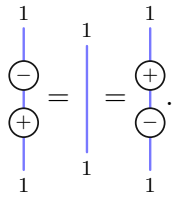


(2C.3)




**Lemma 2C.4.** *We have*

(2C.5)

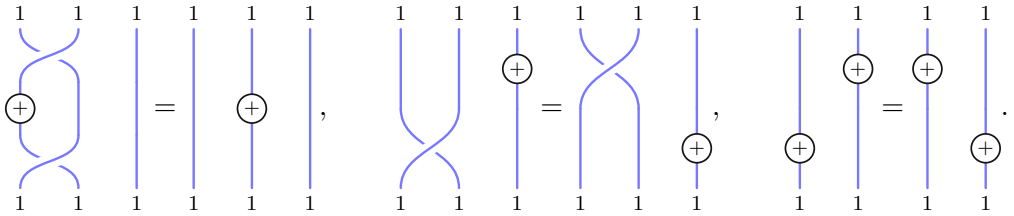



*Proof.* A visualization exercise. □

**Lemma 2C.6.** *Adjoining coils to* Web *together with Equation 2C.5 and isotopies moving splits, merges and crossings across the cut gives* AWeb*. On a thin boundary,*

$$X_iX_k = X_kX_i, \qquad H_iX_iH_i = X_{i+1}, \qquad H_iX_k = X_kH_i \quad (k \neq i, i+1).$$

*In pictures, (with arbitrary unrelated strands in between or to the left and right)*



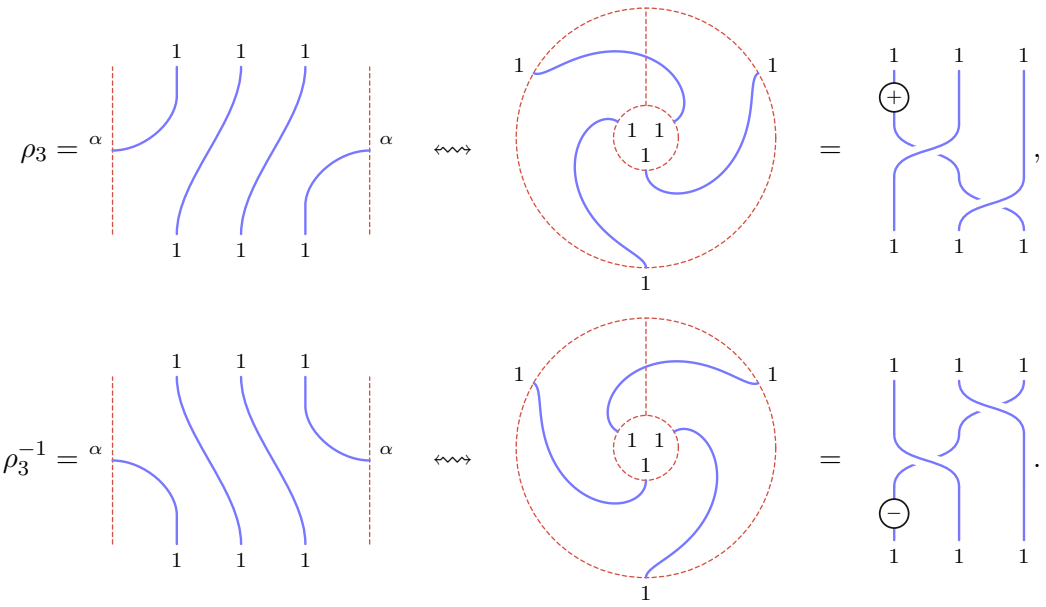

*Strands not involved in a relation remain fixed.*

*Proof.* Use the coil presentation of [LTV23, Section 4A], with the crossing and winding conventions fixed above. In particular, positive rotation here is $\rho^K$ there, and negative rotation is $\rho_K$. The displayed relations follow by moving coils across the cut. To compare with the integral presentation in [SSW25, Definition 2.1], cut the positive winding on an $a$-edge to obtain their solid dot $D_a$. Their hollow dot is its inverse. Moving $D_a$ through a crossing gives $H_{a,b}(D_a \otimes 1_b) = (1_b \otimes D_a)H_{b,a}^{-1}$, as in [SSW25, (2.7)]. Moving $D_{a+b}$ through a split or merge gives $D_a \otimes D_b$, as in [SSW25, (2.8)]. Conversely, these dot relations recover the cylindrical presentation by [MS21, Theorem 2.6], with the opposite braiding since the left tensor factor passes underneath. Splitting an $a$-edge into thin strands, winding them all and merging back gives $[a]!D_a$ by digon removal. This and its negative winding version give [SSW25, (2.9)] over $\mathcal{A}$. □

**Definition 2C.7.** Let $\rho_N$ be the positive cyclic rotation of $\omega_N$ (cf. Equation 2C.8). Under the Hecke realization below, it corresponds to $[2, 3, \ldots, N, N+1]$ in the extended affine symmetric group, not its degree-zero affine subgroup. Write $H_0$ for the crossing through the cut between the last and first thin positions. For $1 \leq a < N$, obtain $\beta_a$ from a reduced word for $w_a = [a+1, \ldots, N, 1, \ldots, a]$ using positive thin crossings. A determinant dot on an $a$-edge denotes the product of its $a$ thin winding variables. ◇

For three thin strands,

(2C.8)



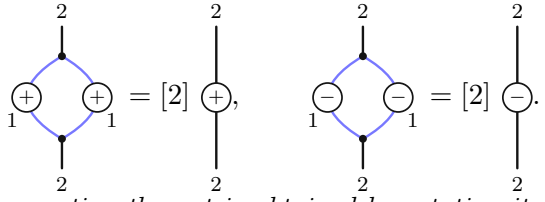

The words are $\rho_3 = X_1H_1H_2$ and $\rho_3^{-1} = H_2^{-1}H_1^{-1}X_1^{-1}$.

**Lemma 2C.9.** *Boundary rotation satisfies*

$$\rho_N^a = (X_1 \cdots X_a)\beta_a, \qquad X_1 = \rho_N H_{N-1}^{-1} \cdots H_1^{-1}. \tag{2C.10}$$

*Also $\rho_N H_i \rho_N^{-1} = H_{i+1}$, with indices modulo $N$. Thus $\beta_a$ moves the last $a$ input positions to the first $a$ output positions, which are then wound once.*

*On thick edges, use block crossings and determinant dots; for thickness two,*




*The thick dots are $X_1X_2$ and its inverse. A vertex meeting the cut is obtained by rotating its edges away from the cut, applying the ordinary vertex, and rotating back.*

*Proof.* Cutting open the rotation gives the braid and winding words. The product of winding variables passes through a split or merge, leaving the digon scalar [2] in the last display. □

*Remark* 2C.11. Annular webs, also called affine webs and other names in the literature, have been studied in various settings; see, e.g., [CK18, FP14, LTV23, Que15, SW25, SSW25]. ◇

2D. **Hecke and Schur categories.** The following is a reinterpretation of, e.g., [Lus89], and many other sources.

**Definition 2D.1.** The extended affine symmetric group $\widehat{S}_N$ consists of bijections $w : \mathbb{Z} \to \mathbb{Z}$ with $w(i+N) = w(i) + N$. The degree of $w$ is $N^{-1}\sum_{i=1}^{N}(w(i) - i)$; the degree-zero subgroup is $W = \widetilde{S}_N$, and $W_0 = \langle s_1, \dots, s_{N-1}\rangle \cong S_N$. The rotation is $\tau = [2, 3, \dots, N, N+1]$.

The extended affine Hecke algebra $\mathcal{H}_{\mathrm{aff}} = \mathcal{H}_{\mathrm{aff}}(N)$ is the $\mathcal{A}$-algebra generated by $H_1, \dots, H_{N-1}$ and $X_1^{\pm 1}, \dots, X_N^{\pm 1}$, subject to the braid, quadratic and Bernstein relations of Lemma 2B.4 and Lemma 2C.6, now read algebraically. Put

$$\tau = X_1H_1 \cdots H_{N-1}, \qquad H_0 = \tau H_{N-1}\tau^{-1}, \qquad q = v^{-2}, \qquad T_i = v^{-1}H_i.$$

The $H_i$ with $1 \le i < N$ generate the finite subalgebra $\mathcal{H}(W_0)$, and $(T_i + 1)(T_i - q) = 0$. Write $H_w = v^{\ell(w)}T_w$ for the standard basis, with $T_\tau = \tau$. The degrees are $\deg H_i = 0$ and $\deg X_i = \deg \tau = 1$; the degree-zero subalgebra is the ordinary affine Hecke algebra. ◇

**Definition 2D.2.** Here and below, our parabolic subgroups are generated by subsets of the affine simple reflections. For a finite parabolic subgroup $P$ with longest element $w_P$, put

$$x_P = \sum_{w \in P} T_w, \qquad C_P = v^{\ell(w_P)}x_P, \qquad h_P = v^{\ell(w_P)} \sum_{w \in P} q^{\ell(w)}.$$

The spherical right module is $\mathcal{M}_P = C_P\mathcal{H}_{\mathrm{aff}} = x_P\mathcal{H}_{\mathrm{aff}}$; its generator satisfies $C_P H_i = v^{-1}C_P$ for $s_i \in P$. For a composition $\gamma = (\gamma_1, \dots, \gamma_k)$ of $N$, let $W_\gamma = S_{\gamma_1} \times \cdots \times S_{\gamma_k} \subset W_0$ permute the consecutive blocks of sizes $\gamma_1, \dots, \gamma_k$. Write $\mathcal{M}_\gamma = \mathcal{M}_{W_\gamma}$, $x_\gamma = x_{W_\gamma}$ and $C_\gamma = C_{W_\gamma}$. Zero parts, when used to pad a boundary, represent empty blocks. ◇

**Definition 2D.3.** The affine Schur category $\mathsf{Schur}_N$ has compositions of $N$ as objects and morphisms $\mathrm{Hom}_{\mathsf{Schur}_N}(\gamma, \alpha) = \mathrm{Hom}_{\mathcal{H}_{\mathrm{aff}}}(\mathcal{M}_\gamma, \mathcal{M}_\alpha)$, with composition of module maps. ◇

**Lemma 2D.4.** *There is an $\mathcal{A}$-linear isomorphism of categories $\Phi_N : \mathsf{AWeb}_N \to \mathsf{Schur}_N$, fixing the boundary compositions. In particular,*

$$\Phi_N : \mathrm{Hom}_{\mathsf{AWeb}}(\gamma, \alpha) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{H}_{\mathrm{aff}}}(\mathcal{M}_\gamma, \mathcal{M}_\alpha). \tag{2D.5}$$

*For finite parabolics $L \subset P$, its split and merge maps are*

$$\iota_{P,L} : \mathcal{M}_P \longrightarrow \mathcal{M}_L, C_P h \longmapsto C_P h, \quad \mu_{L,P} : \mathcal{M}_L \longrightarrow \mathcal{M}_P, C_L h \longmapsto C_P h.$$

*For adjacent edges $(a, b)$, take $P = S_{a+b}$ and $L = S_a \times S_b$; a vertex across the cut is transported by rotation as in Lemma 2C.9.*

*There is an algebra embedding, in fact an isomorphism,*

$$\jmath_N : \mathcal{H}_{\mathrm{aff}} \xrightarrow{\sim} \mathrm{End}_{\mathsf{AWeb}}(\omega_N).$$

*It sends $H_i$ to the thin crossing in positions $i, i+1$, $X_i^{\pm 1}$ to the corresponding winding dot, and $\tau$ to $\rho_N$. Thus $H_{w_a}$ is represented by $\beta_a$. Its finite restriction is $\mathcal{H}(W_0) \cong \mathrm{End}_{\mathsf{Web}}(\omega_N)$.*

*Proof.* With the presentation in Lemma 2C.6, the isomorphism is [SSW25, Theorem 4.16, Proposition 4.18(1)], with $q_{\mathrm{SSW}} = v$ and our flag parameter $q = v^{-2}$. See also [Gre99], [TVW17, Theorem 3.22] and [Bru25]. To fix the vertex normalization, let $P^L$ be the minimal right-coset representatives for $P/L$. Then

$$\sigma_{P/L} = \sum_{x \in P^L} v^{\ell(w_P) - \ell(w_L) - \ell(x)} H_x, \qquad C_P = \sigma_{P/L} C_L = C_L \sigma_{P/L}^*, \qquad H_w^* = H_{w^{-1}}.$$

This gives the split as an inclusion of right ideals. The induced-module presentation makes the merge well-defined. Both maps are integral, and the shuffle expansion gives

$$\mu_{L,P} \circ \iota_{P,L} = \frac{h_P}{h_L}\, \mathrm{id}_{\mathcal{M}_P}, \qquad h_{S_a} = [a]!.$$

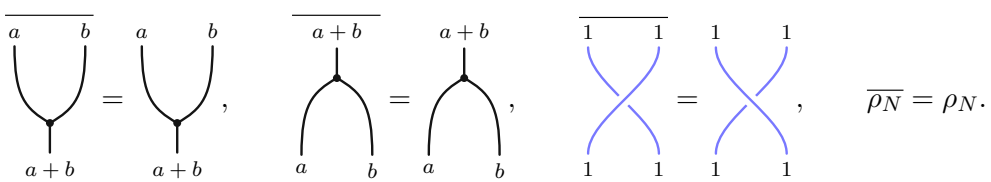
For adjacent edges this is exactly the digon scalar in Equation 2B.2. These are the normalized maps of [TVW17, LT26] and [SSW25, (3.21)–(3.22)]. The thin boundary has $\mathcal{M}_{\omega_N} = \mathcal{H}_{\mathrm{aff}}$. Since these are right modules, $h$ acts by left multiplication $L_h$, and $L_a \circ L_b = L_{ab}$. Apply Equation 2D.5; the finite case is analogous. □

Henceforth we identify Schur maps with webs using $\Phi_N$. The algebraic degree is the signed, thickness-weighted-winding across the cut, so this identification also respects degree.

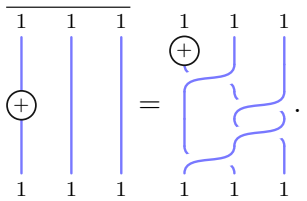
2E. **Standard and canonical bases.** We recall the standard and canonical bases of affine Schur maps, with the normalization needed below. Under the identification in Lemma 2D.4, the latter can be viewed as canonical annular webs.

**Definition 2E.1.** The Hecke bar is the multiplicative involution fixing $\tau$ and sending $v$ to $v^{-1}$ and $H_i$ to $H_i^{-1}$. We use the KL normalization $C_w = v^{\ell(w)} \sum_{y \le w} P_{y,w}(v^{-2}) T_y$. In particular, $C_{w_P} = C_P$ and $C_{s_i} = H_i + v$. On Schur maps the induced bar is $\overline{f}(C_\gamma h) = \overline{f(C_\gamma)} h$. ◇

Thus $C_{s_i} = b_i$ is the merge-split web of Equation 2B.5. Moreover, in pictures,



For thick crossings, the same rule reads $\overline{H_{a,b}} = H_{b,a}^{-1}$. For winding dots, Equation 2C.10 gives $\overline{X_1} = \rho_N H_{N-1} \cdots H_1$. For example, on three thin strands,




The other positions and inverse dots follow from the annular relations.

**Definition 2E.2.** For three-position boundaries $\alpha, \gamma$, double cosets are indexed by periodic matrices $A = (a_{ik})_{i,k\in\mathbb{Z}}$. Their entries are nonnegative, each row and column has finite support, and

$$a_{i+3,k+3} = a_{ik}, \qquad \sum_k a_{ik} = \alpha_i, \qquad \sum_i a_{ik} = \gamma_k.$$

Here the boundaries are extended periodically. Rows record target edges and columns source edges in the unwrapped strip; shifting both indices by three moves to the next copy.

Let $w_{0,\gamma}$ be the longest element of $W_\gamma$ and put $L_\gamma = \ell(w_{0,\gamma})$. For the double coset indexed by $A$, let $d(A)$ and $w_A^+$ be its shortest and longest representatives; thus the coset is $W_\alpha d(A) W_\gamma$. Define $\phi_A(x_\gamma h) = \sum_{w\in W_\alpha d(A) W_\gamma} T_w h$, and the normalized standard and canonical maps by

$$[A]_v = v^{\ell(w_A^+) - L_\gamma} \phi_A, \qquad \theta_A(C_\gamma h) = C_{w_A^+} h \quad (h \in \mathcal{H}_{\mathrm{aff}}).$$

Thus $\theta_A$ is the right $\mathcal{H}_{\mathrm{aff}}$-module map determined by the image of the cyclic generator $C_\gamma$. We order matrices by Bruhat order on their longest representatives, and put

$$\operatorname{dimdeg}(A) = \sum_{\substack{1\le i\le 3,\ k\le i \\ j<l}} a_{ij} a_{kl}.$$

The sum is finite by periodicity and finite support. ◇

**Lemma 2E.3.** *The $[A]_v$ and $\theta_A$ form the standard and canonical bases, respectively. Moreover,* $\operatorname{dimdeg}(A) = \ell(w_A^+) - L_\gamma$, *and the change of basis is*

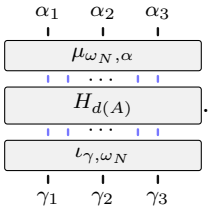
$$\theta_A = \sum_{\substack{Z \\ w_Z^+ \le w_A^+}} v^{\ell(w_A^+) - \ell(w_Z^+)} P_{w_Z^+, w_A^+}(v^{-2}) [Z]_v. \tag{2E.4}$$

*$\theta_A$ is the unique bar invariant element with leading term $[A]_v$ and proper coefficients in $v\mathbb{Z}[v]$.*

*Proof.* The degree formula is [DF15, (2.0.3)] with parameter $v^{-1}$, or [Cui14, Lemma 7.2] with $t = v^{-1}$. Grouping the KL expansion into double cosets gives the change of basis and triangularity [Cui14, Lemma 3.1, Theorem 3.1, (3.4)–(3.5)]. Bar invariance follows from that of $C_{w_A^+}$. □

There is also a diagrammatic description of the standard basis. Write $\iota_{\gamma,\omega_N}$ and $\mu_{\omega_N,\alpha}$ for the total split and merge, and put $I_A = W_\alpha \cap d(A) W_\gamma d(A)^{-1}$. Then

$$h_{I_A} [A]_v = \mu_{\omega_N,\alpha} \circ \jmath_N(H_{d(A)}) \circ \iota_{\gamma,\omega_N} =$$



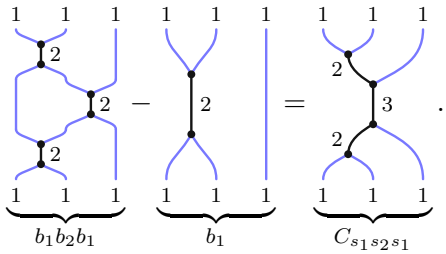

This is a diagrammatic sandwich, in the spirit of sandwich cellularity as in, e.g., [Tub24]. Indeed, parabolic multiplication gives

$$C_\alpha H_{d(A)} C_\gamma = h_{I_A} v^{\ell(w_A^+)} \sum_{w\in W_\alpha d(A) W_\gamma} T_w,$$

and the two sides define the same map on $C_\gamma$. The factor $h_{I_A}$ accounts for repeated symmetrization at the overlap. Canonical webs are bar invariant triangular combinations of these standard webs; they need not be single diagrams.

*Remark* 2E.5. These bases come from [BLM90, Du92, Lus99], in the induced-trivial convention of [Deo87]; see also [McG03] for a useful summary. They differ from the elementary ribbon basis of [SSW25, Theorem 4.16]: a composite web must be expanded in the standard basis before one can read its KL coefficients. An analogous distinction appears in the classical web calculus: Khovanov–Kuperberg (KK) showed that the $\mathfrak{sl}_3$ web basis is not in general dual canonical [KK99]. ◇

Already on three thin strands, a composite web can require a canonical correction:




Indeed, expanding $b_i = H_i + v$ and applying the quadratic relation gives

$$b_1b_2b_1 - b_1 = H_{s_1s_2s_1} + v(H_{s_1s_2} + H_{s_2s_1}) + v^2(H_{s_1} + H_{s_2}) + v^3 = C_{S_3}.$$

The final web is the total merge followed by the total split, which represents $C_{S_3}$ by Lemma 2D.4. Thus even a bar invariant composite need not itself be canonical.

**Definition 2E.6.** Return to $N = 2m + 1$ and put $\lambda = (N, 0, 0)$ and $\nu = (m, 1, m)$. We work in the degree-zero part of

$$\mathcal{C}_m = \mathrm{Hom}_{\mathcal{H}_{\mathrm{aff}}}(x_\lambda \mathcal{H}_{\mathrm{aff}}, x_\nu \mathcal{H}_{\mathrm{aff}}).$$

The zeros pad the source $(N)$; intermediate boundaries may have five edges. For $1 \le j \le m$, define the following matrices. We display rows $1, 2, 3$ and columns $-2, 1, 4$, in that order:

$$A_{j;m} = \begin{pmatrix} 1 & m-j & j-1 \\ 0 & 1 & 0 \\ j-1 & m-j & 1 \end{pmatrix}, \qquad B_m = \begin{pmatrix} 0 & m & 0 \\ 0 & 1 & 0 \\ 0 & m & 0 \end{pmatrix}.$$

All other entries in these rows vanish; extend periodically. The row sums are $\nu$, and the column sums over all rows $i \in \mathbb{Z}$ are $\lambda$, extended periodically. Indeed, the three displayed columns of $A_{j;m}$ sum to $j, N-2j, j$. Since their indices are all congruent to 1 modulo 3, the contributions of their periodic translates add to $N$ in each column congruent to 1 modulo 3; the remaining columns are zero. ◇

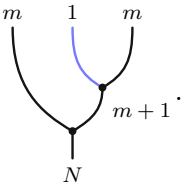
Write $u = u_{j;m}$ and $C = C_{j;m} = uw_0$, with shifts $\delta_i = \delta_i(j)$ as in the introduction.

**Lemma 2E.7.** *The longest representatives of $A_{j;m}$ and $B_m$ are* $w^+_{A_{j;m}} = C = uw_0, \qquad w^+_{B_m} = w_0.$ *Their degrees are*

$$\mathrm{dimdeg}(A_{j;m}) = \ell(u) = 2j(N-j) - 2(m-j) - 3 =: D_{j;m}, \qquad \mathrm{dimdeg}(B_m) = 0.$$

(2E.8) *Finally, the ordinary split through the intermediate edge $m+1$ is both standard and canonical:*

$[B_m]_v = \theta_{B_m} =$




*Proof.* The row blocks for $\nu$ are

$$R^\nu_1 = [1, m], \qquad R^\nu_2 = \{m+1\}, \qquad R^\nu_3 = [m+2, N], \qquad R^\nu_{i+3} = R^\nu_i + N.$$

Define source blocks for $\lambda$ similarly. Relative position is $a_{ik} = |R^\nu_i \cap C(R^\lambda_k)|$. The nonempty source blocks have indices $1 + 3t$ and contain $1 + tN, \dots, N + tN$. A residue $r \in R^\nu_i$ contributes to column $1 - 3\delta_r$. Counting gives $A_{j;m}$; zero shifts give $B_m$.

To identify the longest representatives, note that $u$ has increasing window. Hence $C = uw_0$ has all finite right descents and $\ell(C) = \ell(u) + \ell(w_0)$. If $p(r)$ is the position of $r + N\delta_r$ in the window of $u$, then $C^{-1}(r) = N + 1 - p(r) - N\delta_r$. Within $[1, m]$ and $[m+2, N]$, the shifts are weakly increasing. Equal adjacent shifts give $p(r) < p(r+1)$; increasing shifts make $C^{-1}$ strictly decrease through $-N\delta_r$. Thus every generator of $W_\nu$ is a left descent of $C$. At the omitted finite generators the shifts drop

from 1 to 0 and from 0 to $-1$, so neither is a left descent. The unique element with all prescribed parabolic left and right descents is the longest representative.

It remains to compute the length. In the increasing window inversion formula, negative-zero and zero-positive shift pairs each contribute $(j-1)(N-2j)+(m-j)$, while negative-positive pairs contribute $2j^2-1$. All other pairs contribute zero. Adding gives the asserted length and degrees.

For $B_m$, the canonical map and the iterated split all fix $C_{w_0}$, proving the last claim. □

For a basis vector $b$, write $[b]f$ for its coefficient in $f$, and let $[v]$ extract the coefficient of $v$. A splitter drawn inside the brackets means its coefficient in the normalized standard basis. This denotes coefficient extraction, not composition or closure.

**Proposition 2E.9.** *The required affine KL coefficient is read from the canonical web by*

$$\big[[B_m]_v\big]\theta_{A_{j;m}} = v^{D_{j;m}} P_{w_0,uw_0}(v^{-2}).$$

*In particular,* $[v]\big(\big[[B_m]_v\big]\theta_{A_{j;m}}\big) = \mu_{j;m}$.

*Proof.* Apply Equation 2E.4 with $Z = B_m$. Since $D_{j;m}$ is odd, the term of degree $(D_{j;m}-1)/2$ in the KL polynomial becomes the coefficient of $v$. □

As we will see in Section 3, the standard coefficients of our main web are straightforward to compute, leaving only the base change to the canonical basis (which we carry out in Section 4). Afterwards, Proposition 2E.9 directly yields the required coefficient.

## 3. The main web

The following is the main construction.

3A. **Six split and merge maps.** Fix $m \geq 2$ and $N = 2m+1$. Set

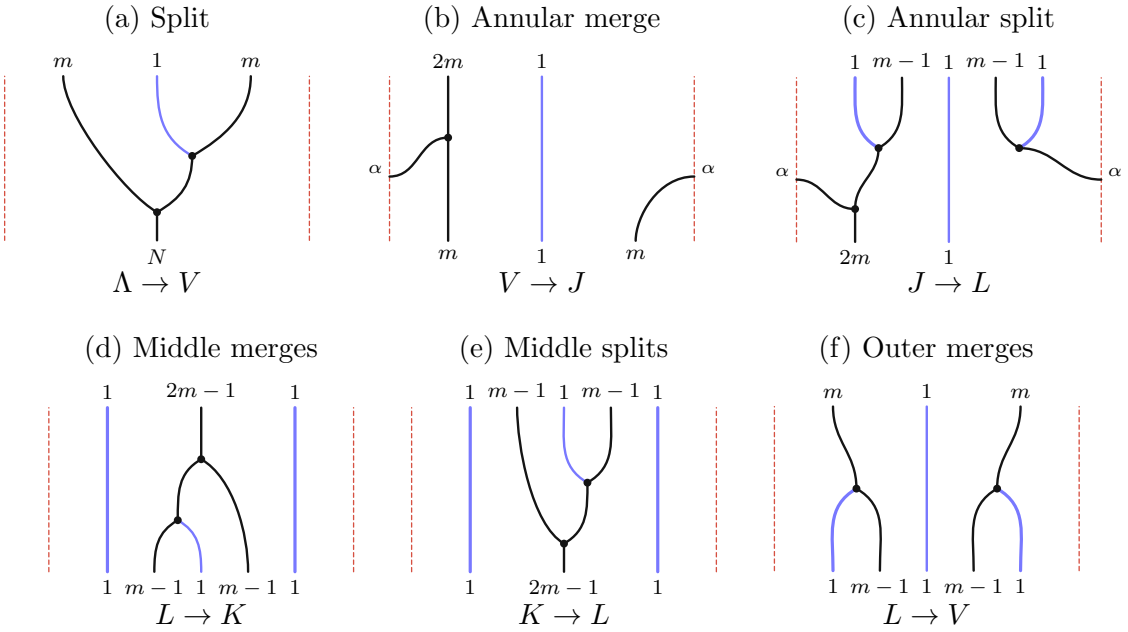
$$\Lambda = (N), \quad V = (m,1,m), \quad J = (2m,1), \quad L = (1,m-1,1,m-1,1), \quad K = (1,N-2,1).$$

The corresponding finite parabolic subgroups are denoted by $P_V, P_J, P_L, P_K$ and specified below; $\Lambda$ corresponds to $W_0$.

The annular web $\mathcal{R}_m$ is the composite

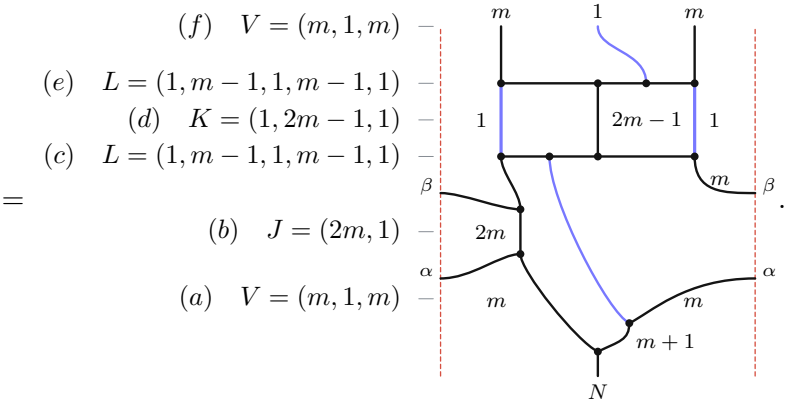
$$\Lambda \xrightarrow{\text{(a)}} V \xrightarrow{\text{(b)}} J \xrightarrow{\text{(c)}} L \xrightarrow{\text{(d)}} K \xrightarrow{\text{(e)}} L \xrightarrow{\text{(f)}} V.$$

The six maps are the following.



(3A.1) Stacking the six maps gives the full web, with the successive boundaries marked on the left:

$\mathcal{R}_m =$




The blue unit strands exposed in (c) pass unchanged through (d)–(e) and rejoin in (f).

3B. **The symmetrizer expression.** By Lemma 2D.4, a split $P \to Q$, $Q \subset P$, is the inclusion $M_P \to M_Q$; the reverse merge sends $C_Q h$ to $C_P h$. After rotating the intermediate boundary $J$ as below, (a)–(f) give the parabolic chain $W_0 \supset P_V \subset P_J \supset P_L \subset P_K \supset P_L \subset P_V$, where $W_0 = \langle s_1, \dots, s_{N-1} \rangle$ and

$$P_J = \langle s_i : i \in \mathbb{Z}/N\mathbb{Z} \setminus \{m, m+1\} \rangle, \quad P_K = \langle s_2, \dots, s_{N-2} \rangle,$$
$$P_V = W_0 \cap P_J, \qquad P_L = P_J \cap P_K = P_K \cap P_V.$$

Their types are $P_V \cong S_m \times S_m$, $P_J \cong S_{2m}$, $P_K \cong S_{2m-1}$ and $P_L \cong S_{m-1} \times S_{m-1}$. The $2m$-edge of $J = (2m, 1)$ crosses the cut, so $P_J$ contains $s_0$ but is finite. In (b), $\tau^m$ moves the last $m$-edge of $V$ to the front, where it merges with the first one. In (c) the split moves an $m$-edge back across the cut. By Lemma 2C.9 the two maps are integral and have degrees $m$ and $-m$, respectively. Their composite has degree zero. Writing $J_{\mathrm{std}} = \langle s_1, \dots, s_{N-2} \rangle$, its middle symmetrizer is

$$\tau^{-m} C_{J_{\mathrm{std}}} \tau^m = C_{P_J} = \tau^{m+1} C_{J_{\mathrm{std}}} \tau^{-(m+1)},$$

since $\tau^N$ is central.

**Lemma 3B.1.** *Identify the right spherical module with $C_{W_0}\mathcal{H}_{\mathrm{aff}}$ by $M_e \mapsto C_{W_0}$, where $M_e H_i = v^{-1} M_e$ for $1 \le i < N$. In the inverse-permutation convention,*

$$\mathcal{R}_m = \frac{M_e C_{P_J} C_{P_K} C_{P_V}}{d_m}, \qquad d_m = h_{P_V} h_{P_L}^2 = [m]!^2 [m-1]!^4. \tag{3B.2}$$

*This quotient is integral and bar invariant.*

*Proof.* Evaluate the six maps of Section 3A on $C_{W_0}$ using Lemma 2D.4. Over $\mathbb{Q}(v)$, a merge from $Q$ to $P$ sends $z$ to $C_P z / h_Q$, since $C_P C_Q = h_Q C_P$. The merges (b), (d) and (f) thus give the value $C_{P_V} C_{P_K} C_{P_J} C_{W_0} / (h_{P_V} h_{P_L}^2)$. The overlaps $W_0 \cap P_J = P_V$, $P_J \cap P_K = P_L$ and $P_K \cap P_V = P_L$ give the three digon factors. Thus the quotient represents the six integral maps, and inversion reverses the product as stated. Every $C_P$ and $h_P$ is bar invariant. □

*Remark* 3B.3. The use of parabolic chains is classical in the Schur algebroid; see [Wil11]. Elias–Ko relate this combinatorics to ordinary type $A$ webs via an associated graded construction [EK23]. For products of parabolic KL elements and their canonical corrections, see also [AS21]. ◇

We now evaluate the Schur maps on $C_{W_0}$ and apply the anti-involution $H_w^* = H_{w^{-1}}$. Write

$$C_A = \big(\theta_A(C_{W_0})\big)^*, \qquad \Delta_A = \big([A]_v(C_{W_0})\big)^*.$$

We retain the original matrix label $A$ for these vectors. This changes the realization, but not the expansion coefficients. Thus $C_B = M_e$ and $[\Delta_A] C_Z = [[A]_v] \theta_Z$. Indeed,

$$[A]_v(C_{W_0}) = v^{\ell(w_A^+)} \sum_{w \in P_V d(A) W_0} T_w,$$

and $*$ replaces this coset by $W_0 d(A)^{-1} P_V$ without changing the scalar. The anti-involution $*$ sends $H_w$ to $H_{w^{-1}}$ and reverses products. The bar sends $H_w$ to $H_{w^{-1}}^{-1}$, sends $v$ to $v^{-1}$, and preserves products.

3C. **Counting flags.** Let $Q$ be a prime power. All flags and lattices counted in this subsection are over $\mathbb{F}_Q$. We use $q$ for the polynomial variable and substitute $q = v^{-2}$ when passing to web coefficients. Schur convolution counts intermediate flags [BLM90, Lus99]. For $\mathcal{R}_m$, we choose a line, a hyperplane and an $m$-space inside an $(m+1)$-space in an $N$-dimensional quotient.

**Definition 3C.1.** Let $A$ be a degree-zero index in the corner of Definition 2E.6, with all nonzero entries in rows $1,2,3$ lying in columns $-2,1,4$. We use the original matrix convention, with output rows and input columns, also for $C_A$ and $\Delta_A$. Degree zero forces the two outer column sums to agree. Set

$$r = \sum_{i=1}^{3} a_{i,-2} = \sum_{i=1}^{3} a_{i,4}, \quad \ell = m - r, \quad d_A = \operatorname{dimdeg}(A),$$
$$a = a_{1,-2}, \quad b = a_{1,-2} + a_{2,-2}, \quad c = m - a_{1,4}, \quad d = m + 1 - a_{1,4} - a_{2,4}.$$

Equivalently, the relative position has $r$ shifts by one period in each direction and $N - 2r$ entries with no shift. Their total shift is zero, which explains the equality above and gives $0 \le r \le m$. In an $N$-dimensional space $E$, fix flags $I_0 \subset K_0$ and $F \subset F'$ with

$$\dim(I_0, K_0) = (r, N-r), \qquad \dim(F, F') = (m, m+1),$$
$$\dim(I_0 \cap F) = a, \quad \dim(I_0 \cap F') = b, \quad \dim(K_0 \cap F) = c, \quad \dim(K_0 \cap F') = d.$$

A gallery is a flag $L \subset I \subset K \subset H$ of dimensions $(1, m, m+1, N-1)$, with $L \subset F$, $F' \subset H$ and $I_0 \subset I \subset K \subset K_0$. ◇

We set $\binom{n}{k}_q = 0$ for $k < 0$ or $k > n$. The choices of $L, H$ split into four cases; their counting polynomials are evaluated at $q = Q$:

| | 0 | 1 |
|---|---|---|
| line $(\epsilon)$ | $L \subset I_0$ | $L \not\subset I_0$ |
| hyperplane $(\eta)$ | $H \supset K_0$ | $H \not\supset K_0$ |

$$\mathsf{L}_0 = (a)_q, \qquad \mathsf{L}_1 = (c)_q - (a)_q,$$
$$\mathsf{H}_0 = (d-\ell-1)_q, \quad \mathsf{H}_1 = (\ell+b)_q - (d-\ell-1)_q.$$

Omit infeasible cases before evaluating the Gaussian. Put

$$F_A(q) = \sum_{\epsilon,\eta \in \{0,1\}} \mathsf{L}_\epsilon \mathsf{H}_\eta \binom{N-2r-\epsilon-\eta}{m-r-\epsilon}_q (m+1-r-\eta)_q. \tag{3C.2}$$

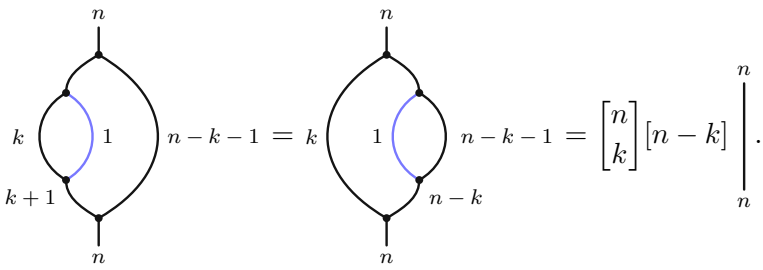
**Lemma 3C.3.** *Over $\mathbb{F}_Q$, the number of galleries is $F_A(Q)$.*

*Proof.* The $\mathsf{L}_\epsilon(Q)$ choices of $L$ and $\mathsf{H}_\eta(Q)$ choices of $H$ are independent since $L \subset F \subset F' \subset H$. The remaining flag $I_0 + L \subset I \subset K \subset K_0 \cap H$ has endpoint dimensions $r+\epsilon$ and $N-r-\eta$:

$$\underset{r+\epsilon}{I_0 + L} \longrightarrow \underset{m}{I} \longrightarrow \underset{m+1}{K} \longrightarrow \underset{N-r-\eta}{K_0 \cap H}$$

Choose $I/(I_0+L)$ in the quotient of dimension $N-2r-\epsilon-\eta$, then the line $K/I$. These give the two remaining factors in Equation 3C.2. □

The same count has a local web calculation. Put $n = N-2r-\epsilon-\eta$ and $k = m-r-\epsilon$ in a feasible case. The quotient flag has dimensions $k, k+1$ in dimension $n$. Choosing the $(k+1)$-space first gives the left web; choosing the $k$-space first gives the right web:




The first equality is associativity and coassociativity; the second removes the two digons. Write $\langle W \rangle$ for the scalar multiplying the straight strand in such an ordinary split–merge web $W$. At $q = v^{-2}$, multiplying the displayed scalar by $v^{-k(n-k)-(n-k-1)}$ gives $\binom{n}{k}_q (n-k)_q$, exactly the two remaining factors in Equation 3C.2. Delete a zero edge at $k = 0$ or $k = n-1$; when $k < 0$ or $k+1 > n$, the flag count is zero.

**Proposition 3C.4.** *Put $D = 2m^2 + 2m - 3$. The standard coefficient of $\mathcal{R}_m$ is*

$$\big[[A]_v\big]\mathcal{R}_m = v^{D-d_A} F_A(v^{-2}).$$

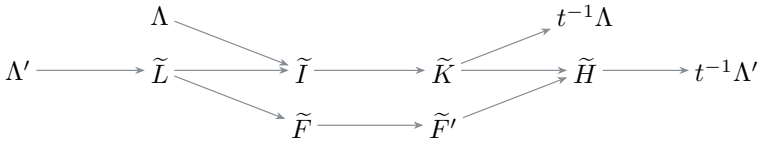
*All standards outside the support of Definition 3C.1 vanish. At the source $B = B_m$,*

$$\big[[B]_v\big]\mathcal{R}_m = v^D (m)_{v^{-2}}^3 \binom{2m-1}{m-1}_{v^{-2}} = v^{m^2}[m]^3 \begin{bmatrix} 2m-1 \\ m-1 \end{bmatrix}. \tag{3C.5}$$

*The coefficient at $A_{m;m}$ is one.*

*Proof.* A lattice is a free rank $N$ $\mathbb{F}_Q[[t]]$-submodule of $\mathbb{F}_Q((t))^N$ spanning that space. A complete periodic chain has $\Lambda_i \subset \Lambda_{i+1}$, $\dim(\Lambda_{i+1}/\Lambda_i) = 1$ and $\Lambda_{i+N} = t^{-1}\Lambda_i$; position $i$ has type $i$ modulo $N$. A chain retains the lattices whose indices belong to a prescribed set of residue classes modulo $N$.

The parabolics $W_0, P_J, P_K, P_V$ fix types $0$; $m, m+1$; $0, 1, N-1$; and $0, m, m+1$, respectively. The convolution counts chains from $\Lambda$ to $(\Lambda', \widetilde{F}, \widetilde{F}')$ satisfying




$$\dim(\widetilde{I}/\Lambda) = m, \quad \dim(\widetilde{K}/\Lambda) = m+1, \quad \dim(\widetilde{L}/\Lambda') = 1, \quad \dim(\widetilde{H}/\Lambda') = N-1.$$

The endpoint lattices have equal determinant degree, so $\dim(\widetilde{I}/\Lambda') = m$ and $\dim(\widetilde{K}/\Lambda') = m+1$. Refining the common facets to complete chambers introduces their Poincaré factors $h_{P_V}, h_{P_L}, h_{P_L}$; forgetting these refinements cancels the denominator in Equation 3B.2.

The inclusions force $\Lambda \subset t^{-1}\Lambda'$ and $\Lambda' \subset t^{-1}\Lambda$. Hence the relative exponents are $-1, 0, 1$, with $r$ of each nonzero sign, proving the support assertion. In $E = t^{-1}\Lambda'/\Lambda'$, set

$$I_0 = (\Lambda + \Lambda')/\Lambda', \qquad K_0 = (t^{-1}\Lambda \cap t^{-1}\Lambda')/\Lambda', \qquad (F, F') = (\widetilde{F}, \widetilde{F}')/\Lambda'.$$

The input chain induces $0 \subset I_0 \subset K_0 \subset E$. Thus columns $-2, 1, 4$ record the flags induced by $F \subset F'$ on $I_0$, $K_0/I_0$ and $E/K_0$, respectively. For instance, $a_{1,-2} = \dim(F \cap I_0)$ and $a_{1,4} = m - \dim(F \cap K_0)$. We therefore get the galleries of Definition 3C.1, counted by $F_A(Q)$. Convolution of orbit characteristic functions counts these intermediate chains, so specialization at $q = Q$ gives the same coefficient as Hecke multiplication [Lus99]. Both the normalized Schur coefficient and $v^{D-d_A}F_A(v^{-2})$ lie in $\mathbb{Z}[v, v^{-1}]$. Their difference vanishes at $v = Q^{-1/2}$ for every prime power $Q$, hence is zero.

For $\ell_P = \ell(w_P)$, we have $\ell_{P_J} = m(2m-1)$, $\ell_{P_K} = (2m-1)(m-1)$, $\ell_{P_V} = m(m-1)$ and $\ell_{P_L} = (m-1)(m-2)$. The normalization exponent is therefore $\ell_{P_J} + \ell_{P_K} + \ell_{P_V} - \ell_{P_V} - 2\ell_{P_L} = D$; the output standard removes $d_A$. At $B$, $r = a = b = 0$, $c = m$, $d = m+1$, and only $(\epsilon, \eta) = (1, 1)$ contributes, giving Equation 3C.5. At $A_{m;m}$ we have $r = m$, $a = c = 1$ and $d = 2$. Thus $I = I_0$, $K = K_0$, $L = F \cap I_0$ and $H = F' + K_0$, so $F_A = 1$ and $d_A = D$. □

Each individual choice is likewise a digon scalar. For $d \geq 1$ and $0 \leq k \leq n$, at $q = v^{-2}$,

$$(d)_q = v^{1-d} \left\langle \; 1 \;\; \overset{d}{\underset{d}{\diamond}} \;\; d-1 \; \right\rangle, \qquad \binom{n}{k}_q = v^{-k(n-k)} \left\langle \; k \;\; \overset{n}{\underset{n}{\diamond}} \;\; n-k \; \right\rangle.$$

At $k = 0, n$, delete the zero edge to obtain an identity. Thus $F_A$ uses products and differences of digon scalars; at $B$,

$$\big[[B]_v\big]\mathcal{R}_m \overset{3}{} = v^{m^2} \left\langle \; 1 \;\; \overset{m}{\underset{m}{\diamond}} \;\; m-1 \; \right\rangle \left\langle \; m-1 \;\; \overset{2m-1}{\underset{2m-1}{\diamond}} \;\; m \; \right\rangle \qquad \big[[B_2]_v\big]\mathcal{R}_2 \overset{3}{} = v^4 \left\langle \; 1 \;\; \overset{2}{\underset{2}{\diamond}} \;\; 1 \; \right\rangle \left\langle \; 1 \;\; \overset{3}{\underset{3}{\diamond}} \;\; 2 \; \right\rangle.$$

For $m = 2$ this is $v^4(v + v^{-1})^3(v^2 + 1 + v^{-2})$, giving $2^3 \cdot 3 = 24$ at $v = 1$. Section 5 expands and corrects this product.

## 4. The base change (main calculation)

The following section is the main calculation, which is a bit tedious, though not difficult. The reader might want to skip to Section 6 for the conclusion, or to Section 5 for an example.

4A. **Parameters and degrees.** Degrees of flag counts restrict the canonical support to six families and the multiplicities of $A_{j;m}$ to zero or one.

**Definition 4A.1.** Write $r = m - \ell$, with $0 \le r \le m$. Every matrix in the support of Proposition 3C.4 has the form

$$A = \begin{pmatrix} a & \ell+s-a & r-s \\ \beta & \delta-\beta & 1-\delta \\ r-a-\beta & \ell+1-s-\delta+a+\beta & s+\delta-1 \end{pmatrix}, \qquad a_{i+3,j+3} = a_{ij}.$$

All entries are nonnegative. Thus $(\beta, \delta) \in \{(0,0),(0,1),(1,1)\}$, and $s \ge 1$ if $\delta = 0$. The intersection dimensions from Definition 3C.1 are $b = a + \beta$, $c = \ell + s$ and $d = \ell + s + \delta$; put $z = s + \delta - 1$. ◇

**Lemma 4A.2.** *For $q = v^{-2}$ and $\ell \ge 1$, the four flag counts give*

$$\begin{aligned} F_A(q) = (\ell)_q((\ell+b)_q - (z)_q)\left((a)_q \binom{2\ell}{\ell}_q + ((c)_q - (a)_q)\binom{2\ell-1}{\ell-1}_q\right) \\ + (\ell+1)_q(z)_q\left((a)_q\binom{2\ell+1}{\ell}_q + ((c)_q-(a)_q)\binom{2\ell}{\ell-1}_q\right). \end{aligned} \tag{4A.3}$$

*For $\ell = 0$, $F_A(q) = (a)_q(d-1)_q$. Empty factors are zero. All summands are nonnegative. Define the length deficit*

$$C = s(\ell+s) + a(\ell+a-s+1) + \beta(\ell+a+\beta) + (\delta-\beta)(s+\delta-1).$$

*Then, with $D = 2m^2 + 2m - 3$, $d_A = 2r(N-r) - C$, $D - d_A = 2\ell(\ell+1) - 3 + C$.*

*Proof.* Substitute the intersection dimensions into Equation 3C.2. Since $c \ge a$ and $\ell + b \ge z$, the differences of the polynomials $(n)_q$ are nonnegative. In the length sum of Definition 2E.2, only two row periods contribute: the predecessor gives $r^2$, the product of the outer column sums, and the current period gives

$$\sum_{1 \le k \le i \le 3} \left(a_{i,-2}a_{k,1} + a_{i,-2}a_{k,4} + a_{i,1}a_{k,4}\right).$$

Their sum is $2r(r+2\ell+1) - C$, giving both length identities. □

4B. **A bound from positivity.** Write val and deg for the least and greatest exponents of a nonzero Laurent polynomial. We apply positivity to $d_m\mathcal{R}_m = M_e C_{P_J} C_{P_K} C_{P_V}$, not to the quotient itself.

**Lemma 4B.1.** *Let $R = \sum_S f_S[S]_v = \sum_S g_S C_S$ be integral and bar invariant. Here $f_S = f_S(v)$ and $g_S = g_S(v)$ are Laurent polynomials. Suppose $d(v) \in \mathbb{N}[v, v^{-1}]$ is nonzero and bar symmetric, and all canonical coefficients of $dR$ are nonnegative. If $g_S \ne 0$, then*

$$\operatorname{val} g_S \ge \operatorname{val} f_S, \qquad \deg g_S \le -\operatorname{val} f_S.$$

*Thus $\operatorname{val} f_S > 0$ excludes $C_S$, and $\operatorname{val} f_S = 0$ makes $g_S$ a nonnegative integer.*

*Proof.* Write $C_T = \sum_S h_{S,T}[S]_v$. These coefficients are nonnegative, with $h_{S,S} = 1$ and $h_{S,T} \in v\mathbb{Z}[v]$ for $S < T$. Hence

$$df_S = dg_S + \sum_{T>S} (dg_T)h_{S,T}$$

is a sum of nonnegative polynomials, so $\operatorname{val}(dg_S) \ge \operatorname{val}(df_S)$. Cancel $\operatorname{val} d$ and use $g_S(v) = g_S(v^{-1})$. If $g_S$ is constant, positivity of $dg_S$ makes it nonnegative. □

For $\mathcal{R}_m = \sum_A M_A(v)C_A$, integrality and bar invariance give $M_A \in \mathbb{Z}[v, v^{-1}]$ and $M_A(v) = M_A(v^{-1})$. Moreover, $d_m M_A \in \mathbb{N}[v, v^{-1}]$: the cleared product embeds as $C_{W_0}C_{P_J}C_{P_K}C_{P_V}$, whose KL coefficients and structure constants are positive [KL80, Spr82]. By Definition 2E.2, $C_A = (\theta_A(C_{W_0}))^* = C_{(w_A^+)^{-1}}$. Thus $d_m M_A$ is exactly the KL coefficient of $C_{(w_A^+)^{-1}}$ in this product.

Lemma 4B.1 and Proposition 3C.4 now give

(4B.2) $\operatorname{val} M_A \geq D - d_A - 2\deg F_A, \quad \deg M_A \leq E_A := 2\deg F_A - (D - d_A)$ whenever $M_A \neq 0$.

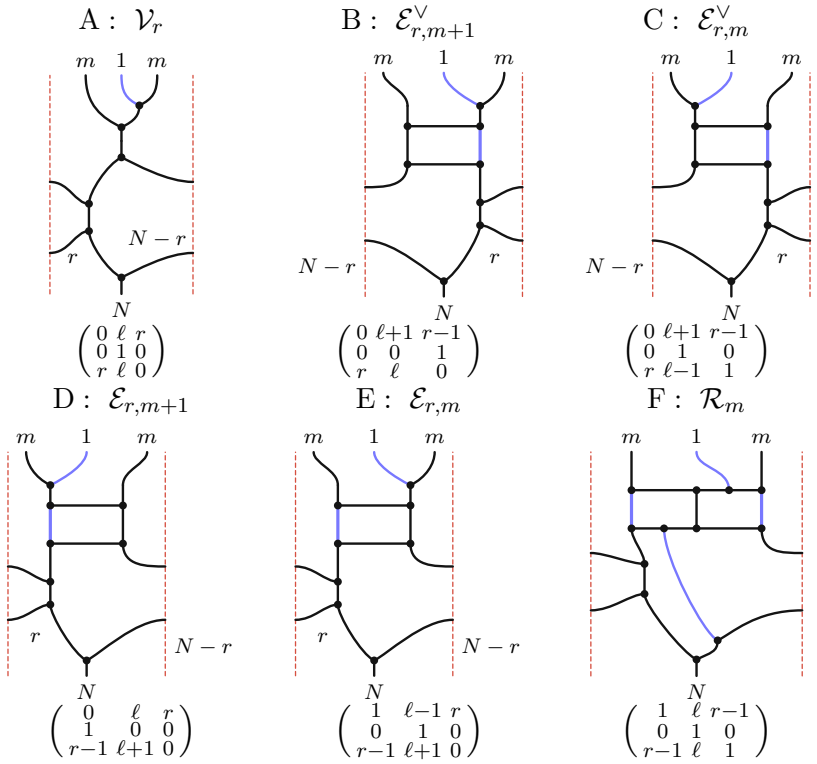
The same positive identity excludes every index whose standard coefficient vanishes.

4C. **The six families.** Families A–F record the surviving web incidences.

**Definition 4C.1.** Family A consists of the full-spherical elements: an endomorphism of $(N)$ followed by the ordinary split to $(m, 1, m)$. Write $Z_r$ for its index at rank $r$. Families E, D retain one line, ending at cuts $m, m+1$ respectively; call them strong and weak. Their duals C, B impose the hyperplane incidence. The operation $\vee$ reflects left–right, exchanges the outer edges and reverses winding. Family F imposes both incidences and consists of the desired $A_{r;m}$ from Definition 2E.6. ◇

The terms indexed by $Z_r$ will give the spherical correction.

The six constructions appear below, with only the first splits labeled in the auxiliary panels. The matrices below the pictures have rows $1, 2, 3$ and columns $-2, 1, 4$, with $\ell = m - r$. The four one-line webs give canonical elements by Lemma 4F.2 and Lemma 4D.1. For family A we take $\mathcal{V}_r - \mathcal{V}_{r-1}$ when $r \geq 1$ (Lemma 4E.4).




**Lemma 4C.2.** *For $\ell \geq 1$, only the following six families can occur in $\mathcal{R}_m = \sum_A M_A(v) C_A$. Here $C_A = (\theta_A(C_{W_0}))^*$ is the canonical vector in the inverse-permutation convention; its coefficient $M_A(v)$ is the same as that of $\theta_A$ in the canonical web expansion. A matrix is feasible when its entries are nonnegative. The last column gives the feasible ranks left by the degree bound, including $\ell = 0$. Every exponent of a nonzero $M_A$ lies in $[-E, E]$, where $E = 2\deg F_A - (D - d_A)$.*

| *family* | $(a, b, c, d)$ | $C$ | $\deg F_A$ | $E$ | *possible ranks* $r$ |
|---|---|---|---|---|---|
| A | $(0, 0, \ell, \ell+1)$ | $0$ | $\ell^2 + 2\ell - 3$ | $2\ell - 3$ | $0 \leq r \leq m-2$ |
| B | $(0, 0, \ell+1, \ell+1)$ | $\ell+1$ | $\ell^2 + 2\ell - 2$ | $\ell - 2$ | $1 \leq r \leq m-2$ |
| C | $(0, 0, \ell+1, \ell+2)$ | $\ell+2$ | $\ell^2 + 2\ell - 1$ | $\ell - 1$ | $1 \leq r \leq m-1$ |
| D | $(0, 1, \ell, \ell+1)$ | $\ell+1$ | $\ell^2 + 2\ell - 2$ | $\ell - 2$ | $1 \leq r \leq m-2$ |
| E | $(1, 1, \ell, \ell+1)$ | $\ell+2$ | $\ell^2 + 2\ell - 1$ | $\ell - 1$ | $1 \leq r \leq m-1$ |
| F | $(1, 1, \ell+1, \ell+2)$ | $2\ell+3$ | $\ell^2 + 2\ell$ | $0$ | $1 \leq r \leq m$ |

*At $\ell = 0$, only the index of $A_{m;m}$ can occur. For every feasible family* F *index, including $A_{m;m}$, $M_A \in \{0, 1\}$; these indices are exactly $A_{r;m}$, $1 \leq r \leq m$.*

*Proof.* Put $f_0 = \ell^2 + 2\ell - 3$. For each nonzero term of Equation 4A.3, call its degree minus $f_0$ its degree increment. The four possible increments are $2a + \beta$, $a + s + \beta$, $a + s + \delta$, $2s + \delta - 1$. By

Equation 4B.2, their maximum must give $E \geq 0$. There are three possible middle rows. *Middle row* $(0,1,0)$*:* $(\beta,\delta) = (0,1)$. Here the largest increment is at most $2\max(a,s)$, except when $a = s \geq 1$, when it is at most $2a+1$. At $a = s = 0$ the third term is empty. With $M = \max(a,s)$ and $u = \min(a,s)$, this bounds $E$ by

$$(2-M-u)\ell - M^2 - u^2 + Mu + 3M - u - 3 + 2\mathbf{1}_{M=u\geq 1}.$$

At $M = 2$, $u = 0,1$ give $-1, -\ell - 1$; on $M = u \geq 2$ the bound is $-2(u-1)\ell - (u-1)^2$. It decreases with $M$, losing the extra 2 off the diagonal. Thus $M \geq 2$ is impossible, leaving $a, s \in \{0,1\}$ and families $\mathrm{A}, C, E, F$.

*Middle row* $(0,0,1)$*:* $(\beta,\delta) = (0,0)$. Here we have $s \geq 1$ and the largest increment is at most $\max(2a, 2s-1)$. If $a \geq s$, then $E \leq (2-a-s)\ell - a^2 - s^2 + as + 3a - 3$. At $a = s$ the right side is $(2-2s)\ell - s^2 + 3s - 3 < 0$, and it decreases as $a$ increases. If $s \geq a+1$, then $E \leq (2-a-s)\ell - a^2 - s^2 + as + 4s - a - 5$. For $a = 0, s \geq 2$ it starts at $-1$ and decreases; for $a \geq 1$, it starts at $(1-2a)\ell - (a-1)^2 - 1 < 0$ when $s = a+1$ and decreases. Only $a = 0, s = 1$ remains: family B.

*Middle row* $(1,0,0)$*:* $(\beta,\delta) = (1,1)$. Here the largest increment is at most $\max(2a+1, 2s)$. If $a \geq s$, then $E \leq (1-a-s)\ell - a^2 - s^2 + as + 2a - 2$. At $s = 0, a = 1$ it is $-1$; at $a = s \geq 1$ it is $(1-2a)\ell - (a-1)^2 - 1$. Both decrease with $a$. If $s \geq a+1$, $E \leq (1-a-s)\ell - a^2 - s^2 + as - 2a + 4s - 4$. At $a = 0, s = 1$ it is $-1$. For $a \geq 1, s = a+1$ it is $-2a\ell - a^2 + a - 1 < 0$. Increasing $s$ decreases it. Only $a = s = 0$ remains, giving family D. Substituting these six positions into the nonempty terms gives the displayed degrees exactly.

*The boundary case* $\ell = 0$. A nonzero coefficient has $a \geq 1$ and $d \geq 2$, and $\deg F_A = a + d - 3$. Feasibility forces $s = a$ in case $(0,1)$, and $s \in \{a, a+1\}$ in the other two cases. The respective values of $E$ are

| $(\beta,\delta)$ | $s = a$ | $s = a+1$ |
|---|---|---|
| $(0,1)$ | $-(a-1)^2$ | infeasible |
| $(0,0)$ | $-a^2 + 3a - 3$ | $-(a-1)^2 - 1$ |
| $(1,1)$ | $-(a-1)^2 - 1$ | $-a^2 + a - 1$. |

Only $a = s = 1$, $(\beta,\delta) = (0,1)$ has nonnegative excess.

In family F, the unique top-degree term is $(\ell+1)_q \binom{2\ell+1}{\ell}_q$, monic; at $\ell = 0$, $F_A = 1$. Hence $[\Delta_A]\mathcal{R}_m$ has valuation zero and constant one, so $M_A$ is constant. Since $d_m M_A$ is nonnegative and $d_m$ has lowest coefficient one, the positive identity above Equation 4B.2 gives $0 \leq M_A \leq 1$. The displayed family F matrix is $A_{r;m}$ by Definition 2E.6. At $r = m$ its uncorrected leading coefficient is one and it is the unique maximal standard index, so in fact $M_{A_{m;m}} = 1$. □

4D. **Output refinement.** We have the following lemma.

**Lemma 4D.1.** *For finite output parabolics $P \subset T$, partition a coarse double coset $D$ into fine double cosets $E$. Then*

$$\Delta_D^T = \sum_{E \subset D} v^{\ell(w_D^+) - \ell(w_E^+)} \Delta_E^P. \tag{4D.2}$$

*A coarse canonical vector $C_Z$ remains canonical after refinement. Write $h_{D,Z}^T = [\Delta_D^T]C_Z$ and $h_{E,Z}^P = [\Delta_E^P]C_Z$. Then, for $E \subset D$, $h_{E,Z}^P = v^{\ell(w_D^+) - \ell(w_E^+)} h_{D,Z}^T$. At the origin $Z_0$, provided $T \subseteq W_0$, the only fine coset is $B$ and the factor is one, so the distinguished coefficient is unchanged.*

*Proof.* Partition the unnormalized $T$-sum and compare normalization factors. For $E \subset D$, we have $w_E^+ \leq w_D^+$, with equality exactly for the fine coset containing $w_D^+$. Refining a coarse canonical vector therefore gives one leading fine standard with coefficient one; all other coefficients lie in $v\mathbb{Z}[v]$ and have smaller longest representatives. The underlying Hecke vector remains bar invariant, proving canonicality. If $T \subseteq W_0$, then $W_0 T = W_0 P = W_0$, with longest representative $w_0$. □

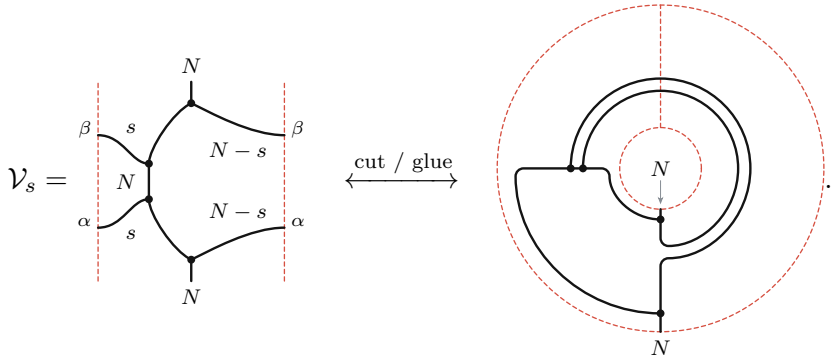
4E. **No retained line.** Next, we need the following.

**Definition 4E.1.** For arbitrary $N$ and $0 \leq s \leq \lfloor N/2 \rfloor$, put $J_s = \langle s_i : i \neq s \rangle \cong S_N$ and $I_s = W_0 \cap J_s \cong S_s \times S_{N-s}$. Define

$$\mathcal{V}_s = \frac{M_e C_{J_s} C_{W_0}}{h_{I_s}^2}, \qquad \mathcal{V}_0 = M_e.$$

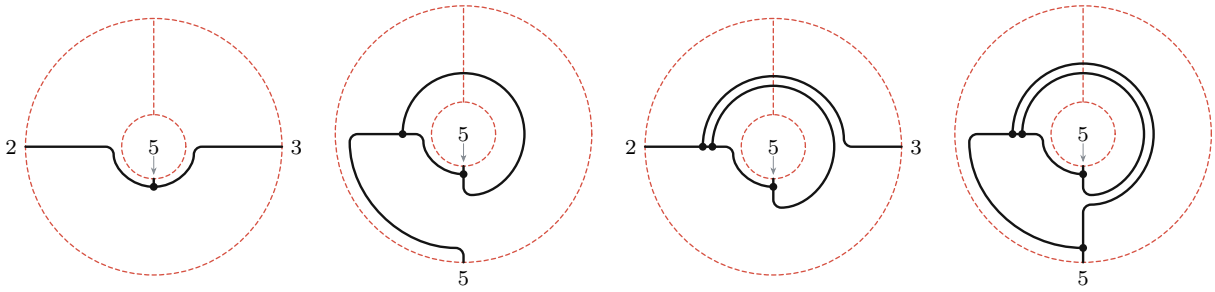
The web $\mathcal{V}_s$ splits $(N)$ into $(s, N-s)$, merges and splits across the cut, then merges ordinarily. The spherical standard $Z_r$ has $r$ exponents of each sign and $N-2r$ zeros, with length gap $2r(N-r)$. We keep the notation after output refinement. ◇

Matching $\alpha, \beta$ across the cuts gives




For $\mathcal{V}_2 : (5) \to (5)$, the annular stages are

(a) Split $(2,3)$ (b) Annular merge (c) Annular split (d) Merge




All inner inputs have label 5; (b) merges the entire edge.

**Lemma 4E.2.** *The standard coefficient of $\mathcal{V}_s$ at $Z_r$ is zero for $r > s$, and for $0 \le r \le s$ equals*

$$v^{2s(N-s)-2r(N-r)} \binom{N-2r}{s-r}_{v^{-2}}. \tag{4E.3}$$

*Moreover, in the canonical basis, $\mathcal{V}_s = \sum_{a=0}^{s} C_{Z_a}$.*

*Proof.* The gallery chooses a type-$s$ lattice $\widetilde{U}$ between the fixed vertices:

$$\Lambda + \Lambda' \subset \widetilde{U} \subset t^{-1}\Lambda \cap t^{-1}\Lambda', \qquad \dim(\widetilde{U}/\Lambda) = s.$$

At rank $r$, this chooses an $(s-r)$-space in dimension $N-2r$, giving the Gaussian. Normalization gives $2(\ell(w_0) - \ell(w_{I_s})) - 2r(N-r) = 2s(N-s) - 2r(N-r)$. Thus the standard coefficients are integral. The symmetrizer expression for $\mathcal{V}_s$ is bar invariant.

Apply Lemma 4B.1 with $d = h_{I_s}^2$; clearing $d$ gives a positive product of canonical symmetrizers. The standard coefficients have valuation zero and constant one, so the degree bound and bar symmetry make all multiplicities constant. Descending through ranks, larger canonical terms have positive degree at smaller standards. The constant term therefore gives each multiplicity as one. □

Put $h = s - r$ and $n = N - 2r$. Since $2s(N-s) - 2r(N-r) = 2h(n-h)$, the coefficient is

$$\big[[Z_r]_v\big]\mathcal{V}_s = v^{h(n-h)} \left\langle \; h \; \begin{array}{c} n \\ \bigcirc \\ n \end{array} \; n-h \; \right\rangle.$$

The digon has extreme degrees $\pm h(n-h)$ with coefficients one, hence valuation zero and constant one after shifting.

**Lemma 4E.4.** *The spherical canonical elements are*

$$C_{Z_0} = \mathcal{V}_0 = M_e, \qquad C_{Z_s} = \mathcal{V}_s - \mathcal{V}_{s-1} \quad (s \ge 1).$$

*For $0 \leq r \leq s$, put $n = N - 2r$, $h = s - r$, and $q = v^{-2}$. Then*

$$[[Z_r]_v]C_{Z_s} = v^{2s(N-s)-2r(N-r)} \left( \binom{n}{h}_q - q^{n-2h+1} \binom{n}{h-1}_q \right). \tag{4E.5}$$

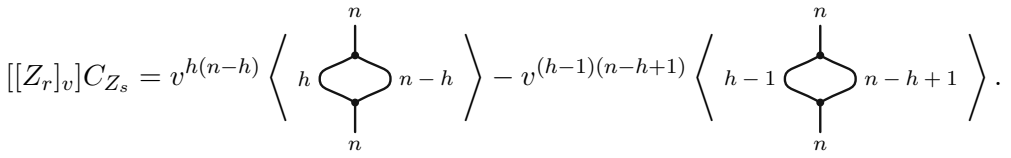
*Proof.* Subtract the canonical expansions in Lemma 4E.2, then the coefficients in Equation 4E.3. Their normalization exponents differ by $2(N - 2s + 1) = 2(n - 2h + 1)$, giving the power of $q$. The case $s = 0$ is immediate. □

Write $K_{\alpha,\beta}(t)$ for the Kostka–Foulkes transition polynomial from Schur to Hall–Littlewood functions. Gaussian reciprocity with $t = v^2$ rewrites Equation 4E.5 as

$$[[Z_r]_v]C_{Z_s} = \binom{n}{h}_t - \binom{n}{h-1}_t = K_{(2^h,1^{n-2h}),(1^n)}(t).$$

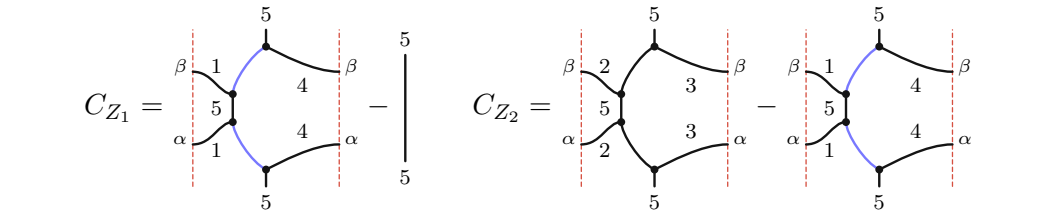
Here a Gaussian with lower index $-1$ is zero. The last equality is the two-column case of Macdonald's hook formula [Mac95, Chapter III, Section 6, Example 2]. This spherical affine KL interpretation goes back to Lusztig [Lus83] and Kato [Kat82]; [NR03, Theorem 3.4(d)] gives an explicit account of the inverse parameter convention. This gives the full-spherical correction in our three-output calculation. In the web calculus, for $h \geq 1$ the same coordinate is

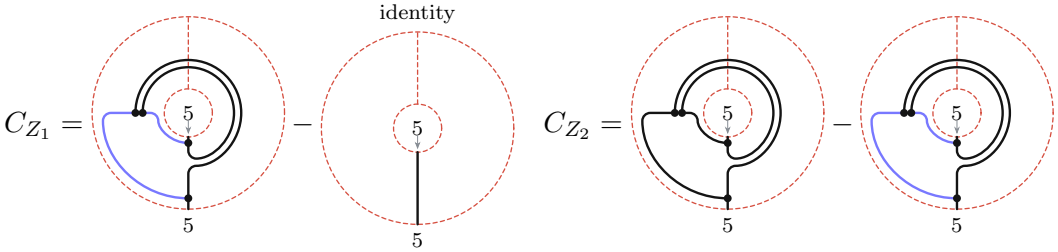
$$[[Z_r]_v]C_{Z_s} = v^{h(n-h)} \left\langle \; h \;\; n-h \; \right\rangle - v^{(h-1)(n-h+1)} \left\langle \; h-1 \;\; n-h+1 \; \right\rangle .$$




This is an identity of scalar coordinates: remove each digon, then take the difference. At $h = 1$ the second web is the straight $n$-strand, and at $h = 0$ the coordinate is one.

For $N = 5$, cut open and on the annulus, these differences are



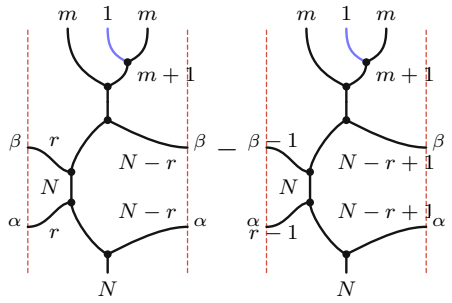




Here $\mathcal{V}_0$ is the identity; the first splits are $(1, 4)$ and $(2, 3)$.

At $N = 5$, the coefficients at the origin are

| web | $[[Z_0]_v]$web | canonical difference at $Z_0$ |
|---|---|---|
| $\mathcal{V}_0$ | $1$ | $1$ |
| $\mathcal{V}_1$ | $1 + v^2 + v^4 + v^6 + v^8$ | $v^2 + v^4 + v^6 + v^8$ |
| $\mathcal{V}_2$ | $1 + v^2 + 2v^4 + 2v^6 + 2v^8 + v^{10} + v^{12}$ | $v^4 + v^6 + v^8 + v^{10} + v^{12}$ |

(4E.6) Appending the splitter of Equation 2E.8 gives




$(r \geq 1)$.

At rank zero the refined canonical web is the ordinary splitter $B_m$.

4F. **One retained line.** We need the following.

**Definition 4F.1.** Let $j \geq 1$, $N \geq 2j+1$ and $j \leq f \leq N-j$, and put

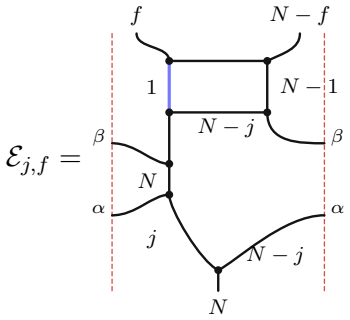
$$J_j = \langle s_i : i \neq j \rangle, \quad K_1 = \langle s_2, \dots, s_{N-1} \rangle, \quad T_f = \langle s_i : 1 \leq i < N,\ i \neq f \rangle.$$

The web below represents $\mathcal{E}_{j,f} : (N) \to (f, N-f)$, defined by

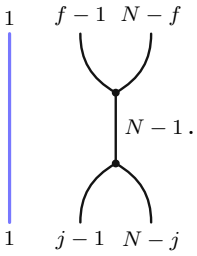
$$\mathcal{E}_{j,f} = \frac{M_e C_{J_j} C_{K_1} C_{T_f}}{[j]![N-j]!\,[j-1]![N-j]!\,[f-1]![N-f]!}.$$

◇

The web and its middle stage are

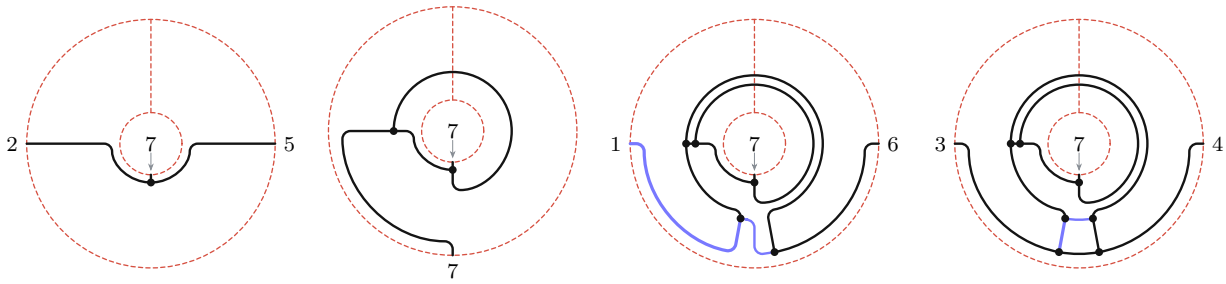



with middle stage




At $j = 1$ or $f = 1$, delete the zero edge and its identity vertex. For $\mathcal{E}_{2,3} : (7) \to (3,4)$, the annular stages are

(a) Split $(2,5)$ (b) Annular merge (c) Retain one line (d) Split and rejoin




At (c) the output is $(1,6)$; at (d) split the 6 into $(2,4)$ and join the blue line to the 2-edge.

For a rank $r$ standard, set $n = N - 2r$, $s = j - r$, $d = f - r$. Its intersection dimensions $a = \dim(I_0 \cap F)$ and $c = \dim(K_0 \cap F) = d + b$ give the nonnegative matrix

$$Z = \begin{pmatrix} a & d+b-a & r-b \\ r-a & n-d-b+a & b \end{pmatrix}.$$

The length deficit and normalization exponent are

$$C = b(d+b) + a(n-d+a-b), \qquad d_Z = 2r(N-r) - C, \qquad E = (2j-1)(N-j) + f - 1.$$

**Lemma 4F.2.** *The coefficient of the normalized standard $Z$ in $\mathcal{E}_{j,f}$ is*

$$v^{E-d_Z}\left((a)_q \binom{n}{s}_q + ((c)_q - (a)_q)\binom{n-1}{s-1}_q\right), \qquad q = v^{-2}. \tag{4F.3}$$

*All other coefficients vanish. The web is a canonical basis element. In particular its coefficient at $B$ is $[[B]_v]\mathcal{E}_{j,f} = v^E (f)_q \binom{N-1}{j-1}_q$.*

*Proof.* A gallery chooses a type-$j$ lattice $\widetilde{U}$, a target vertex, a line $L \subset \widetilde{U}/\Lambda'$, and the prescribed $f$-space $F$ containing $L$. In the finite quotient, these choices are

$$L \subset F \cap K_0, \qquad I_0 + L \subset U \subset K_0, \qquad \dim U = j.$$

The choices with $L \subset I_0$ have counting polynomial $(a)_q \binom{n}{s}_q$, while those with $L \not\subset I_0$ have counting polynomial $((c)_q - (a)_q)\binom{n-1}{s-1}_q$. Successive parabolic intersections are $S_j \times S_{N-j}$, $S_{j-1} \times S_{N-j}$, and

$S_{f-1} \times S_{N-f}$. Their Poincaré factors give the denominator, and subtracting their longest lengths from the numerator lengths gives $E$. This proves the integral coefficient formula; the symmetrizer expression is bar invariant.

For canonicality, the length gap and the degrees of the two nonzero polynomial pieces are

$$E - d_Z = 2s(n-s) - (n-d-s+1) + C,$$
$$f_1 = a - 1 + s(n-s), \qquad f_2 = c - 1 + (s-1)(n-s).$$

Subtracting gives

$$2f_1 - (E - d_Z) = -(a-1)(n-d+a-1) + b(a-d-b) - s,$$
$$2f_2 - (E - d_Z) = -(n-d) + s - 1 + b(2-d-b) - a(n-d+a-b).$$

For the first piece $a \geq 1$, $n - d \geq s$, and $a \leq d + b$. Its bound is nonpositive, with equality only at $s = 0, a = 1, b = 0$ when $d \geq 1$, or $s = 0, a = b = 1$ when $d = 0$. In each case $r = j$. With $n = N - 2j$ and $d = f - j$, the leading standard has coefficient one and matrix

$$\begin{pmatrix} 1 & d-1 & j \\ j-1 & n-d+1 & 0 \end{pmatrix} \quad (d \geq 1), \qquad \begin{pmatrix} 1 & 0 & j-1 \\ j-1 & n & 1 \end{pmatrix} \quad (d = 0).$$

For the second piece, if $d \geq 1$ then $b(2-d-b) \leq 0$, so the bound is at most $-1$. If $d = 0$, the hypotheses force $r = j = f$ and $s = 0$, while $n \geq 1$; since $b(2-b) \leq 1$, the bound is at most $-n < 0$. Thus every proper standard coefficient lies in $v\mathbb{Z}[v]$. Any other Bruhat-maximal index would have a nonzero bar invariant coefficient in $v\mathbb{Z}[v]$, impossible. Thus the leading index is the unique maximum, proving canonicality. □

For $N = 5$, refine the output to $(2, 1, 2)$. The distinguished standard coefficient is unchanged by this refinement. At $B$, the normalization exponent is $E - (f-1) - (j-1)(N-j) = j(N-j)$. For $j = 1$ the second digon is an identity, so the general formula and its first example are

$$[[B]_v]\mathcal{E}_{j,f} = v^{j(N-j)} \left\langle \; 1 \;\; f \;\; f-1 \; \right\rangle \left\langle \; j-1 \;\; N-1 \;\; N-j \; \right\rangle$$

$$[[B_2]_v]\mathcal{E}_{1,2} = v^4 \left\langle \; 1 \;\; 2 \;\; 1 \; \right\rangle = v^3 + v^5.$$

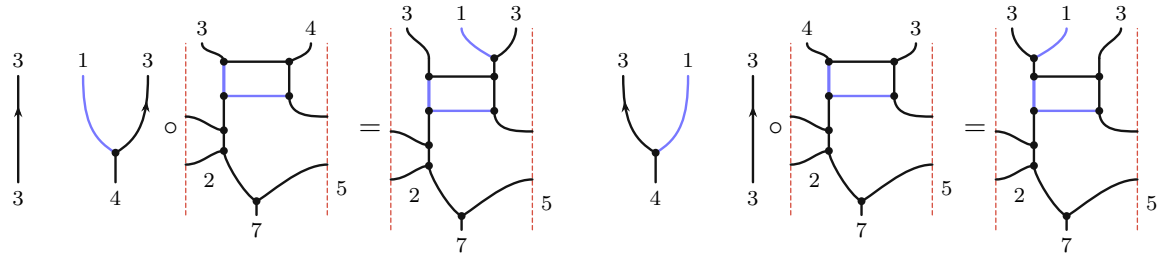
Reflection gives the same coefficient for $\mathcal{E}^{\vee}_{1,2}$.

4G. **The refined family webs.** The strong and weak one-line refinements are

$$(m, m+1) \longrightarrow (m, 1, m) : \quad m \;\; 1 \;\; m \;\; / \;\; m \;\; m+1, \qquad (m+1, m) \longrightarrow (m, 1, m) : \quad m \;\; 1 \;\; m \;\; / \;\; m+1 \;\; m.$$

To identify the leading matrices, put $j = r$ and $\ell = m - r \geq 1$. Joining the last two rows of family E gives the leading matrix for $f = m$ in Lemma 4F.2. Joining the first two rows of family D gives the one for $f = m + 1$. The length deficits before and after refinement are $\ell + 2$ for E and $\ell + 1$ for D. Thus both refinement exponents in Equation 4D.2 are zero. Reversing both rows and columns gives C from E and B from D.

For $N = 7$, $j = 2$, split the right 4-edge of $\mathcal{E}_{2,3}$ or the left 4-edge of $\mathcal{E}_{2,4}$. Both retain the blue line and end at $(3, 1, 3)$:

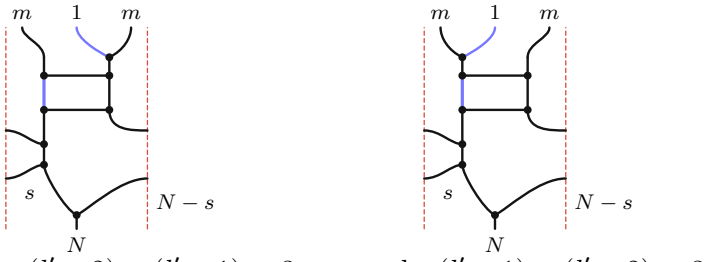


4H. **The one-line degree bound.** Set $N = 2m + 1$. The image families have $f = m$ or $m + 1$ for strong or weak incidence; reflection gives the kernel families. By Equation 4D.2, output refinement to $(m, 1, m)$ preserves their canonical vectors.

**Lemma 4H.1.** *Let a one-line canonical term have rank $s$, and put $l' = m - s$. At a full-spherical standard of rank $r < s$, its minimum degree is $l' + 2$ for the strong family and $l' + 1$ for the weak family. After multiplying by its coefficient in $\mathcal{R}_m$, the minimum degree is at least three in either case. The same holds for the dual families.*

*Proof.* At $Z_r$, both the two-row and three-row matrices have length deficit zero, so Equation 4D.2 introduces no power of $v$. At this standard $a = 0, c = f - r$, so the polynomial in Equation 4F.3 is $(f-r)_q \binom{N-2r-1}{s-r-1}_q$. Subtracting twice its degree from the length gap gives $N - s - f + 1$. The relevance bounds allow multiplicity degrees down to $-(l' - 1)$ and $-(l' - 2)$ respectively. Both resulting margins are three. For $s \le r$ the coefficient vanishes. □

Diagrammatically, the margins are




strong: $(l' + 2) - (l' - 1) = 3$ weak: $(l' + 1) - (l' - 2) = 3$

Reflection gives the other two cases.

4I. **Multiplicity one.** We now control the multiplicity.

**Lemma 4I.1.** *Every family* F *index has multiplicity $M_{A_{j;m}} = 1$ for $1 \le j \le m$.*

*Proof.* The case $j = m$ is already settled in Lemma 4C.2: its family F index is maximal, and all one-line ranks are at most $m - 1$. Hence assume $j < m$. Fix $j$ and an auxiliary rank $s \ge j$. Put $h = s - j$, $k = m - j$ and $\ell = m - s = k - h$. At $A_{j;m}$ the length deficit is $2k + 3$ by Lemma 4C.2. Merging all three rows gives deficit zero, so full-spherical refinement contributes $v^{2k+3}$ by Equation 4D.2. The coarse canonical coefficient lies in $\mathbb{Z}[v]$.

For a strong one-line term (mark dimension $f = m$), the relative parameters in Equation 4F.3 are $n = 2k + 1$, $a = b = 1$, $c = k + 1$, $d = k$. For the weak term ($f = m + 1$), only $c, d$ change, to $k + 2, k + 1$. The codimension statistic is $2k + 2$ for either two-row matrix, so refinement to the three-row matrix contributes $v^{(2k+3)-(2k+2)} = v$. Their counting polynomials are, respectively,

$$\binom{n}{h}_q + ((k+1)_q - 1)\binom{n-1}{h-1}_q, \qquad \binom{n}{h}_q + ((k+2)_q - 1)\binom{n-1}{h-1}_q.$$

The first Gaussian has degree $h(n - h)$. For $h \ge 1$, the second terms have degrees $(h - 1)(n - h) + k$ and $(h - 1)(n - h) + k + 1$, respectively. Their deficits from the first are $(n - h) - k = \ell + 1$ and $(n - h) - (k + 1) = \ell$. For $h = 0$ the second terms vanish. Here $\ell \ge 1$ by Lemma 4C.2. Subtract twice the Gaussian degree from the refined length gap and use the multiplicity bounds in Lemma 4C.2. For $A = A_{j;m}$, write $h_{A,S}$ for the coefficient of $\Delta_A$ in $C_S$. The nonzero contributions have the following lower bounds.

| $S$ | full-spherical | strong | weak |
|---|---|---|---|
| $\operatorname{val} h_{A,S}$ | $2k + 3$ | $\ell + 2h + 1$ | $\ell + 2h + 2$ |
| $\operatorname{val} M_S$ | $3 - 2\ell$ | $1 - \ell$ | $2 - \ell$ |
| $\operatorname{val}(M_S h_{A,S})$ | $2h + 6$ | $2h + 2$ | $2h + 4$ |

Reflection gives the same bounds for the kernel families; the auxiliary canonical coefficients vanish at rank $j > s$.

The remaining terms $C_{A_{s;m}}$ have constant multiplicity zero or one. For $s > j$, their coefficients at $A_{j;m}$ lie in $v\mathbb{Z}[v]$ by strict triangularity; for $s < j$, their smaller length excludes them. Thus only $M_{A_{j;m}} C_{A_{j;m}}$ contributes to the constant one, proving $M_{A_{j;m}} = 1$. □

The constant term at $A_{j;m}$ determines a multiplicity. The later coefficient test at $B_m$ reads the row sum after the spherical correction.

4J. **A Gaussian cancellation.** At a full-spherical standard of rank $r$, put $l = m - r$ and

$$D_l = 2l(l+1) - 3, \qquad e_l = (D_l - 1)/2,$$

$$Y_l(q) = (l)_q^3 \binom{2l-1}{l-1}_q, \qquad R_{n,h}(q) = \binom{n}{h}_q - q^{n-2h+1}\binom{n}{h-1}_q.$$

To remove the nonpositive powers of $v$ from $v^{D_l}Y_l(v^{-2})$, we use the bar-symmetric polynomials

$$g_l(v) = \sum_{t=0}^{l-2} \binom{t+3}{3}\left(v^{2l-3-2t} + v^{-(2l-3-2t)}\right) \quad (l \geq 2), \qquad g_1 = 0.$$

For example, $g_2 = v + v^{-1}$ and $g_3 = v^3 + 4v + 4v^{-1} + v^{-3}$. Write

$$H_l(q) = \sum_{t=0}^{l-2} \binom{t+3}{3}(q^t + q^{2l-3-t}), \qquad M_l(q) = q^{l^2}H_l(q), \qquad H_1 = M_1 = 0.$$

Then $v^{D_l}M_l(v^{-2}) = g_l(v)$. The coefficients $\binom{t+3}{3}$ are those of $(1-q)^{-4}$, whose truncation we use next.

**Lemma 4J.1.** *Fix $m \geq 2$ and put $\Delta_r(q) = \binom{2m+1}{r}_q - \binom{2m+1}{r-1}_q$. As formal power series modulo $q^m$,*

$$q^r H_{m-r}(q) \equiv \frac{q^r}{(1-q)^4} - \binom{m-r+1}{2}q^{m-1} \qquad (0 \leq r \leq m-1),$$

$$\sum_{r=0}^{m-1} q^r \Delta_r(q) \equiv \frac{1-q}{(q;q)_\infty},$$

*where $(q;q)_r = \prod_{i=1}^{r}(1-q^i)$ and $(q;q)_\infty = \prod_{i\geq 1}(1-q^i)$.*

*Proof.* Put $l = m - r$. The coefficients of $H_l$ agree with $(1-q)^{-4}$ through degree $l-2$. At degree $l-1$ the difference is $\binom{l+2}{3} - \binom{l+1}{3} = \binom{l+1}{2}$. This also gives the endpoint $l = 1$, since $H_1 = 0$.

For the second identity, telescope the Gaussian differences:

$$\sum_{r=0}^{m-1} q^r \Delta_r \equiv (1-q)\sum_{r=0}^{m-1} q^r \binom{2m+1}{r}_q \equiv (1-q)\sum_{r\geq 0}\frac{q^r}{(q;q)_r} \pmod{q^m}.$$

Here $\binom{2m+1}{r}_q \equiv 1/(q;q)_r \pmod{q^m}$ for $0 \leq r \leq m-1$. Euler's partition identity $\sum_{r\geq 0} q^r/(q;q)_r = 1/(q;q)_\infty$ finishes the proof [Mac95, Chapter I, Section 2, Example 4]. □

**Lemma 4J.2.** *For every $m \geq 2$, the polynomial*

$$Q_m(q) = Y_m(q) - \sum_{r=0}^{m-2} M_{m-r}(q)R_{2m+1,r}(q)$$

*has degree at most $e_m$ and $[q^{e_m}]Q_m(q) = \binom{m+1}{2}$.*

*Proof.* The unwanted low powers of $v$ correspond to high powers of $q$, which we bring to degree zero by reversing the polynomials. Put $T = m^2 + 2m - 3 = \deg Y_m$. Gaussian reciprocity and reciprocity of $H_l$ give

$$q^T(M_{m-r}R_{2m+1,r})(q^{-1}) = q^r H_{m-r}(q)\Delta_r(q).$$

Every reversed term is a polynomial, and $Y_m$ is reciprocal. We compute the reversal of $Q_m$ modulo $q^m$. In Lemma 4J.1, $\Delta_0(0) = 1$ and $\Delta_r(0) = 0$ for $r > 0$. Since $H_1 = 0$, we may include $r = m-1$. The lemma then gives

$$\sum_{r=0}^{m-1} q^r H_{m-r}\Delta_r \equiv \frac{1}{(1-q)^3(q;q)_\infty} - \binom{m+1}{2}q^{m-1} \pmod{q^m}.$$

On the other hand, $Y_m \equiv ((1-q)^3(q;q)_\infty)^{-1} \pmod{q^m}$. The common series cancels, leaving

$$q^T Q_m(q^{-1}) \equiv \binom{m+1}{2}q^{m-1} \pmod{q^m}.$$

Consequently all terms of $Q_m$ above degree $T - (m-1) = e_m$ vanish, and the coefficient in that degree is $\binom{m+1}{2}$. □

4K. **Removing the full-spherical terms.** Next up:

**Lemma 4K.1.** *The coefficient of $C_{Z_r}$ in $\mathcal{R}_m$ is $g_{m-r}(v)$ for $0 \le r \le m-2$; no full-spherical term of rank $m-1$ occurs. The integral bar invariant web*

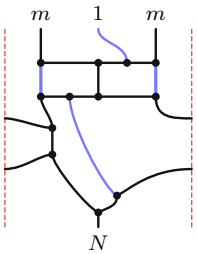
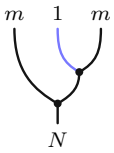
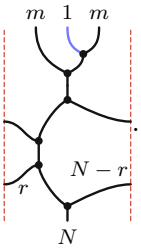
$$\mathcal{R}_m^{\mathrm{crit}} = \mathcal{R}_m - \sum_{r=0}^{m-2} g_{m-r}(v) C_{Z_r}$$

*has at $B$ valuation at least one and coefficient $\binom{m+1}{2}$ at $v$.*

*Proof.* Descend through the spherical ranks. At $Z_r$, put $l = m - r$. The raw coordinate is $v^{D_l} Y_l(v^{-2})$, and the coordinate of $C_{Z_{r+h}}$ is $v^{2h(2l+1-h)} R_{2l+1,h}(v^{-2})$. The one-line contributions have valuation at least three by Lemma 4H.1. The family F multiplicities are constant, and their coefficients here are proper canonical coefficients, hence have positive degree. By Lemma 4E.2, spherical terms of rank less than $r$ have zero $Z_r$-coordinate. Thus, after subtracting the known higher-rank spherical terms, the nonpositive part of the $Z_r$-coordinate is exactly that of $M_{Z_r}$.

The classification excludes $l = 1$. Suppose the multiplicities at larger ranks are known. Since $D_l - D_{l-h} = 2h(2l+1-h)$, subtracting those terms and the proposed $g_l C_{Z_r}$ leaves, at $Z_r$, $v^{D_l} Q_l(v^{-2}) \in v\mathbb{Z}[v]$ by Lemma 4J.2. Hence $M_{Z_r} - g_l \in v\mathbb{Z}[v]$. This difference is bar invariant, so it is zero. At $r = 0$, the same lemma gives valuation at least one and coefficient $\binom{m+1}{2}$ at $v$. □

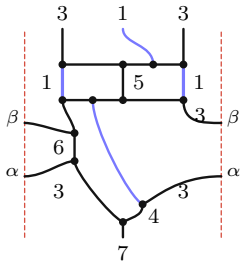
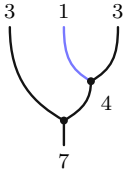
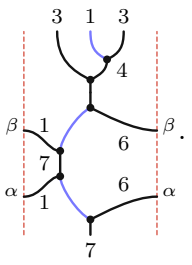
The correction is a linear combination of webs built from splits and merges. Telescoping $C_{Z_r} = \mathcal{V}_r - \mathcal{V}_{r-1}$, with the outputs refined and $g_1 = 0$, gives

$\mathcal{R}_m^{\mathrm{crit}} =$  $- (g_m - g_{m-1})$  $- \sum_{r=1}^{m-2} (g_{m-r} - g_{m-r-1})$ 

All terms have boundary $(N) \to (m, 1, m)$; the sum is empty for $m = 2$. Using Equation 4E.6, the first corrections are

$$\mathcal{R}_2^{\mathrm{crit}} = \mathcal{R}_2 - (v + v^{-1}) C_{Z_0},$$
$$\mathcal{R}_3^{\mathrm{crit}} = \mathcal{R}_3 - (v^3 + 4v + 4v^{-1} + v^{-3}) C_{Z_0} - (v + v^{-1}) C_{Z_1}.$$

For $m = 3$, substitute $C_{Z_0} = B_3$ and $C_{Z_1} = \mathcal{V}_1 - B_3$, with outputs refined, and collect terms:

$\mathcal{R}_3^{\mathrm{crit}} =$  $- \big(g_3(v) - g_2(v)\big)$  $- g_2(v)$ 

At the chosen standard coordinate,

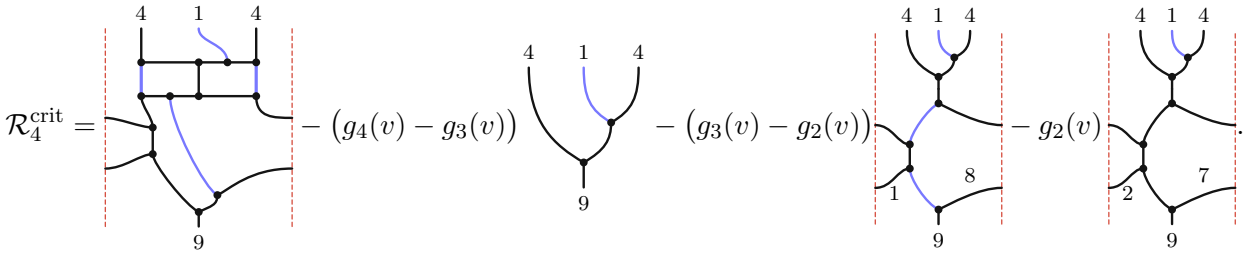
| term | $\mathcal{R}_3$ | $g_3 C_{Z_0}$ | $g_2 C_{Z_1}$ | $\mathcal{R}_3^{\mathrm{crit}}$ |
|---|---|---|---|---|
| $[v]\,[[B_3]_v]$ | 11 | 4 | 1 | $11 - 4 - 1 = 6.$ |

The spherical contribution 1 in this table is itself a digon calculation:

$$[v]\, g_2(v) \left( v^6 \left\langle \; 1 \; \begin{array}{c} 7 \\ \bigcirc \\ 7 \end{array} \; 6 \; \right\rangle - 1 \right) = [v]\,(v + v^{-1})(v^2 + v^4 + \cdots + v^{12}) = 1.$$

Thus $11 - 4 - 1 = 6$ is $\mu_{1;3} + \mu_{2;3} + \mu_{3;3}$, since the one-line terms start in degree three.

For $m = 4$, the boundary is $(9) \to (4, 1, 4)$. Here the refined $C_{Z_1} = \mathcal{V}_1 - B_4$ and $C_{Z_2} = \mathcal{V}_2 - \mathcal{V}_1$ have bottom splits $(1, 8)$ and $(2, 7)$. Collecting terms gives




The coefficient of $v$ sums the four mu-coefficients:

| term | $\mathcal{R}_4$ | $g_4 C_{Z_0}$ | $g_3 C_{Z_1}$ | $g_2 C_{Z_2}$ | $\mathcal{R}_4^{\mathrm{crit}}$ |
|---|---|---|---|---|---|
| $[v]\,[[B_4]_v]$ | 25 | 10 | 5 | 0 | $25 - 10 - 5 - 0 = 10$ |

4L. **The critical coefficient.** The following is our main result.

**Theorem 4L.1.** *For every $m \geq 2$, the endpoints $w_0$ and $C_{j;m}$ satisfy*

$$\sum_{j=1}^{m} \mu_{j;m} = \binom{m+1}{2}. \tag{4L.2}$$

*In particular* $\max_{1 \leq j \leq m} \mu_{j;m} \geq \lceil (m+1)/2 \rceil$.

*Proof.* The six-family classification and the multiplicity-one lemma give

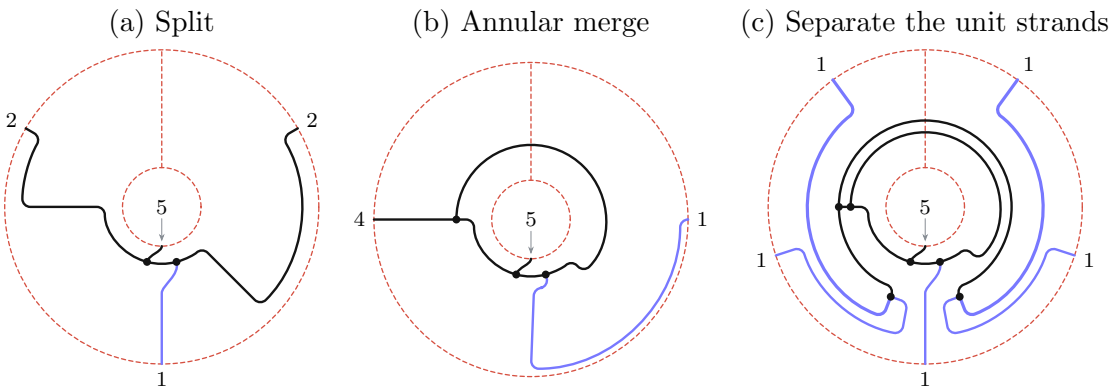
$$\mathcal{R}_m^{\mathrm{crit}} = \sum_{j=1}^{m} C_{A_{j;m}} + \sum_{\text{one-line } S} M_S(v) C_S.$$

The last sum has minimum degree at least three at $B$. Taking $[v]$ and using the canonical-to-standard dictionary gives Equation 4L.2. Its consequence follows by averaging and integrality. □

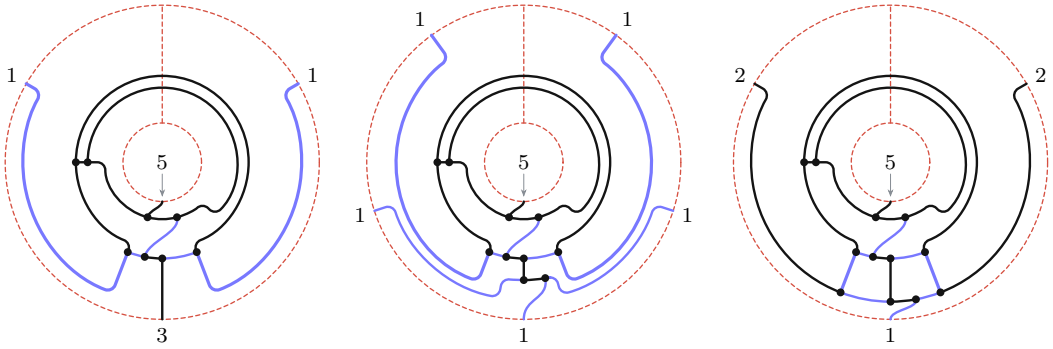
## 5. The first example in pictures

For $m = 2$, the boundary is $(5) \to (2, 1, 2)$ and the intermediate refinement is thin. We compute the two coefficients $\mu_{1;2} = 1$ and $\mu_{2;2} = 2$ directly from the web expansion.

5A. **Constructing and expanding the web.** The completed webs $\mathcal{R}_2 : (5) \to (2, 1, 2)$ and $\mathcal{R}_3 : (7) \to (3, 1, 3)$ appear in Figure 3; their interior labels are in Equation 3A.1. Here are all six stages for $m = 2$. Each composite has input 5; the blue strands bypass the middle merge and split, then rejoin at (f):



(d) Merge the middle (e) Split the middle (f) Rejoin the outer edges



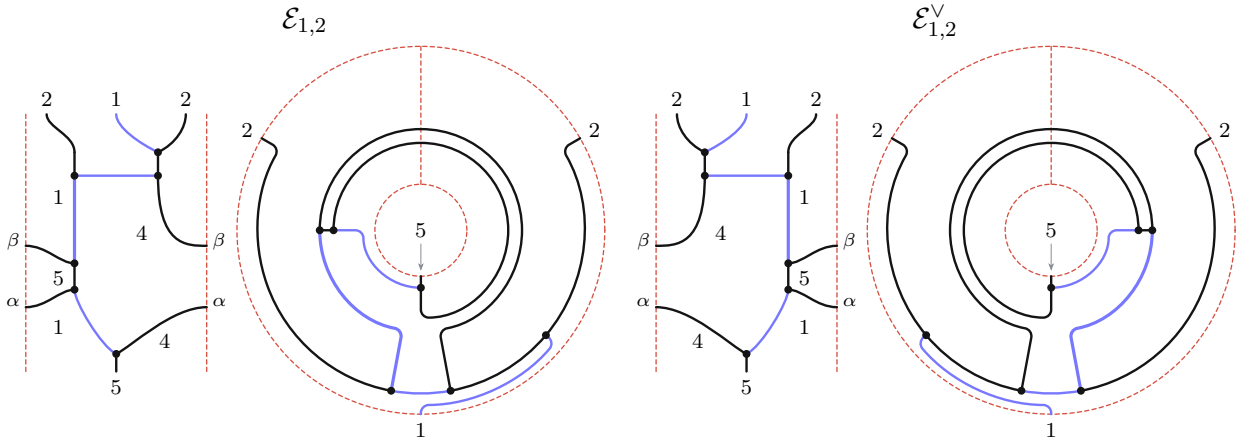

**Lemma 5A.1.** *The canonical expansion is*

$$\mathcal{R}_2 = \theta_{A_{2;2}} + \theta_{A_{1;2}} + \mathcal{E}_{1,2} + \mathcal{E}^{\vee}_{1,2} + (v + v^{-1})\theta_{B_2}. \tag{5A.2}$$

*Here both one-line webs have their outputs refined to* $(2, 1, 2)$.

*Proof.* The classification leaves these five terms. The family multiplicities are one by Lemma 4I.1; the one-line multiplicities are constant with leading coefficient one. Finally, $g_2 = v + v^{-1}$. □

Each pair below shows one of the two one-line webs, cut open and on the annulus:




Removing the source term gives $\mathcal{R}_2^{\mathrm{crit}} = \mathcal{R}_2 - (v + v^{-1})B_2$, as drawn in Figure 5. To isolate $\theta_{A_{2;2}}$, we also need the smaller family term.

5B. **The rank-one correction.** The smaller family element comes from three local maps $b_i = H_i + v$. For a spherical standard $M_w$ indexed by a minimal left $W_0$-coset representative,

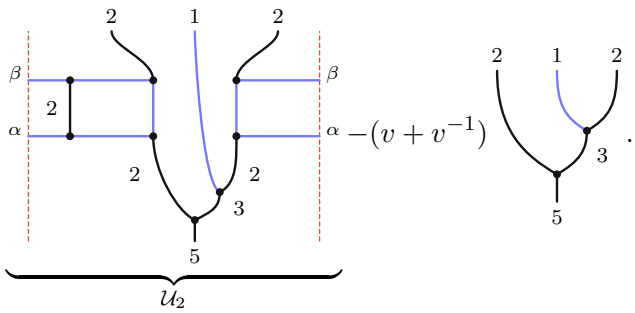
$$M_w b_i = \begin{cases} M_{ws_i} + vM_w, & ws_i \text{ is minimal and } \ell(ws_i) = \ell(w) + 1, \\ M_{ws_i} + v^{-1}M_w, & ws_i \text{ is minimal and } \ell(ws_i) = \ell(w) - 1, \\ (v + v^{-1})M_w, & ws_i \text{ is not minimal.} \end{cases}$$

The last case retains the symmetric input. The product at $0, 4, 1$ gives $\mathcal{U}_2 \longleftrightarrow M_e b_0 b_4 b_1$. Indeed, $M_e b_0 = M_{s_0} + vM_e$. The four terms of $M_{s_0} b_4 b_1$ form the single normalized standard at $A_{1;2}$, while $b_4$ and $b_1$ each act on $M_e$ by $v + v^{-1}$. Thus

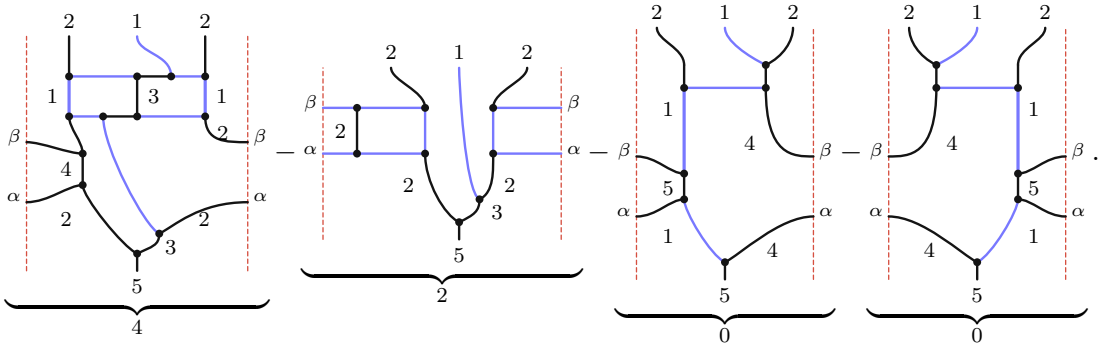
$$\mathcal{U}_2 = [A_{1;2}]_v + (v^{-1} + 2v + v^3)[B_2]_v.$$

To draw $\mathcal{U}_2$, split into five thin strands, apply $b_0$ across the cut, and merge the outer pairs. Matching $\alpha, \beta$ recovers the two branches of $b_0$; the final merges give $b_4 b_1$ in the inverse module. The only nonpositive power in a proper standard coefficient is $v^{-1}$ at the origin. Subtracting $(v + v^{-1})M_e$ leaves all proper coefficients in $v\mathbb{Z}[v]$, so bar invariance gives

$$\theta_{A_{1;2}} = M_e b_0 b_4 b_1 - (v + v^{-1})M_e, \qquad \big[[B_2]_v\big]\theta_{A_{1;2}} = v + v^3.$$

In pictures, $\theta_{A_{1;2}} =$  $-(v+v^{-1})$  .

5C. **Reading the coefficient.** Substitution cancels the source terms in Equation 5A.2: $\theta_{A_{2;2}} = \mathcal{R}_2 - \mathcal{U}_2 - \mathcal{E}_{1,2} - \mathcal{E}^\vee_{1,2}$. All four webs have boundary $(5) \to (2,1,2)$. The numbers underneath are their $[v]\,[[B_2]_v]$ coordinates:

$\theta_{A_{2;2}} =$  .

Their $B_2$ coordinates are

| web | coefficient of $[B_2]_v$ |
|---|---|
| $\mathcal{R}_2$ | $v^{-1}+4v+7v^3+7v^5+4v^7+v^9$ |
| $\mathcal{U}_2$ | $v^{-1}+2v+v^3$ |
| $\mathcal{E}_{1,2}$ | $v^3+v^5$ |
| $\mathcal{E}^\vee_{1,2}$ | $v^3+v^5$ |

Subtracting the last three rows from the first yields

$$\big[[B_2]_v\big]\theta_{A_{2;2}} = 2v+4v^3+5v^5+4v^7+v^9.$$

For $u_{2;2} = (-4,-1,3,7,10)$, the endpoint $C_{2;2} = u_{2;2}w_0$ has length gap nine. Thus $P_{w_0,C_{2;2}}(q) = 1+4q+5q^2+4q^3+2q^4$. One gets $\mu_{2;2} = 4-2 = 2$; the smaller element has $\mu_{1;2} = 1$.

## 6. Application to Guralnick's conjecture

We now pass to highest weights and finite group cohomology. Algebraic group extensions and cohomology are taken among rational modules. The input $(N)$ gives $M_e \mapsto C_{w_0}$, so the lower endpoint corresponds to the trivial module.

6A. **The highest weights.** Fix $N = 2m+1$ and let $k = \overline{\mathbb{F}}_p$, with $p > N$. Write $G = \mathrm{SL}_N(k)$ and let $\varpi_1, \dots, \varpi_{N-1}$ be its fundamental weights. Put $\eta_i = \varpi_i + \varpi_{N-i}$ for $1 \le i \le m$, and $\eta_0 = 0$. For $1 \le j \le m$, set $c = m-j$ and define

$$\lambda_{j;m}(p) = (c+2)\eta_{j-1} + (p-c-3)\eta_j + \eta_m. \tag{6A.1}$$

Write $L(\lambda)$ for the simple rational $G$-module of highest weight $\lambda$.

In sum-zero coordinates, put $\rho_i = (N+1-2i)/2$. For $\xi \in \mathbb{Z}^N$ with $\sum_i \xi_i = 0$, the translation $t_\xi$ has window $t_\xi(i) = i + N\xi_i$ and positive $p$-dot action $(t_\xi z)\cdot_p \lambda = z(\lambda+\rho) + p\xi - \rho, \quad z \in S_N$.

**Lemma 6A.2.** *The weight $\lambda_{j;m}(p) = u_{j;m}^{-1}\cdot_p 0$ is nonzero, dominant, $p$-restricted, $p$-regular and in the root lattice, with sum-zero coordinates*

$$\big(p^{\times(j-1)},\ p-c-2,\ 1^{\times c},\ 0,\ (-1)^{\times c},\ c+2-p,\ (-p)^{\times(j-1)}\big), \tag{6A.3}$$

*where an exponent $\times a$ means $a$ repetitions.*

*Proof.* Write $u = u_{j;m}$ and $b = u^{-1}$. The increasing window is $u(i) = z_i + Ns_i$, with $1 \le z_i \le N$; thus $b$ has permutation part $z^{-1}$ and translation coordinates $-s_i$. Hence $(b\cdot_p 0)_i = -ps_i + \rho_{z_i} - \rho_i = -ps_i + i - z_i$. The negative-shift residues are $(1, \dots, j-1, m+2)$, the positive-shift residues are $(m, N-j+2, \dots, N)$, both in increasing order, and the remaining shifts are zero. Substitution gives Equation 6A.3; adjacent differences give Equation 6A.1.

For $j < m$, the nonzero fundamental-weight coefficients $c+2$, $p-c-3$, 1 lie in $[0, p-1]$; for $j = m$, the weight is $2\eta_{m-1} + (p-2)\eta_m$, again dominant and restricted. The nonzero integral sum-zero vector lies in the root lattice. Finally, $\lambda + \rho$ is congruent modulo $p$ to a permutation of $\rho$, whose coordinate differences have absolute value between 1 and $N - 1 < p$. Thus $\lambda$ is $p$-regular. □

6B. **The cohomology coefficient.** For the KL-to-cohomology identification, assume Lusztig's character formula for the $p$-restricted weights of $G$. For the finite group comparison, also require $p \ge 5(N-1)$. For fixed $N$, all sufficiently large primes satisfy both conditions [AJS94]; see [Lüb20, Theorem 3.1]. We use only the existence of the character-formula-threshold; no explicit bound for it is needed here.

**Lemma 6B.1.** *For these primes and every $1 \le j \le m$,*

$$\dim_k H^1\Big(G, L\big(\lambda_{j;m}(p)\big)\Big) = \dim_k H^1\Big(\mathrm{SL}_N(\mathbb{F}_p), L\big(\lambda_{j;m}(p)\big)\Big) = \mu_{j;m}.$$

*The restricted finite group module is nontrivial and absolutely irreducible.*

*Proof.* Put $b = u_{j;m}^{-1}$, so $\lambda_{j;m}(p) = b \cdot_p 0$ by Lemma 6A.2. Since $u_{j;m}$ is right finite-minimal, $b$ is left finite-minimal. By Section 2, its normalized spherical coefficient at $M_e$ is

$$m_{e,b}(v) = v^{D_{j;m}} P_{w_0, w_0 b}(v^{-2}) = v^{D_{j;m}} P_{w_0, C_{j;m}}(v^{-2}) = \big[[B_m]_v\big]\theta_{A_{j;m}}.$$

Here $w_0 b = (u_{j;m} w_0)^{-1} = C_{j;m}^{-1}$, and simultaneous inversion of both KL indices identifies $P_{w_0, w_0 b}$ with $P_{w_0, C_{j;m}}$. Thus $[v]m_{e,b}(v) = \mu_{j;m}$. Andersen's cohomological interpretation [And86], as in [Lüb20, Theorem 4.5], gives

$$[v]m_{e,b} = \dim_k \mathrm{Ext}^1_G\big(k, L(b \cdot_p 0)\big) = \dim_k H^1\big(G, L(b \cdot_p 0)\big).$$

To see the conventions directly in [CPS09, Theorems 5.4(b) and 6.8(a)], take $-2\rho$ in their negative fundamental alcove. The weights 0 and $b \cdot_p 0$ then have labels $w_0$ and $bw_0$. Conjugation by $w_0$ takes their simple reflections to ours and their two labels to $w_0$ and $w_0 b$. The Weyl module at 0 is $k$. At the regular restricted weight $b \cdot_p 0$ of Lemma 6A.2, the character formula identifies their reduced costandard module with $L(b \cdot_p 0)$ by [CPS09, (1.5.6)].

The Coxeter number is $N$, so Bendel–Nakano–Pillen comparison [BNP01, Theorem 7.5], in the form [Lüb20, Theorem 4.6], applies for $p \ge 5(N-1)$, cohomological bound $s = 1$, base weight $\nu = 0$ and field exponent $r = 1$, giving the finite group equality. Steinberg's restricted-module theorem [Ste63, Theorems 1.3, 6.1 and 7.4] gives irreducibility on restriction; see also [Lüb20, Proposition 5.3]. The nonzero restricted weight gives nontriviality. Since $k$ is algebraically closed, irreducibility is absolute. □

6C. **Passage to the finite simple quotient.** We now wrap up.

**Lemma 6C.1.** *The restricted module $L(\lambda_{j;m}(p))$ descends to a faithful absolutely irreducible module $M_{j;m}(p)$ for $H = \mathrm{PSL}_N(\mathbb{F}_p)$, and*

$$H^1\big(H, M_{j;m}(p)\big) \cong H^1\Big(\mathrm{SL}_N(\mathbb{F}_p), L\big(\lambda_{j;m}(p)\big)\Big).$$

*Proof.* The root-lattice condition makes the algebraic center act trivially. Write $Z$ for the finite center. Its order $\gcd(N, p-1)$ is prime to $p$, so $H^1(Z, L(\lambda_{j;m}(p))) = 0$ by averaging. Inflation–restriction therefore gives the displayed isomorphism. Descent preserves absolute irreducibility and nontriviality. Since $N \ge 5$, the group $H$ is nonabelian simple; its nontrivial representation is therefore faithful. □

**Corollary 6C.2.** *For each $m \ge 2$ and every sufficiently large prime $p$, the group $H = \mathrm{PSL}_{2m+1}(\mathbb{F}_p)$ has the faithful absolutely irreducible modules $M_{j;m}(p)$ defined above, and*

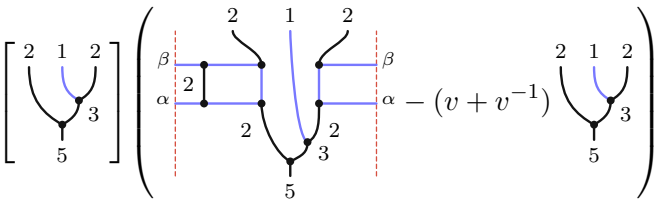
$$\sum_{j=1}^{m} \dim_k H^1\big(H, M_{j;m}(p)\big) = \binom{m+1}{2}, \qquad \max_{1 \le j \le m} \dim_k H^1\big(H, M_{j;m}(p)\big) \ge \left\lceil \frac{m+1}{2} \right\rceil.$$

*Thus Guralnick's absolute boundedness conjecture is false.*

*Proof.* Transfer identifies the summands with $\mu_{j;m}$. Apply Theorem 4L.1, average and use integrality; the lower bound tends to infinity. □

For $m = 2$, the Dynkin coordinates of $\lambda_{1;2}(p)$ and $\lambda_{2;2}(p)$ are their coefficients in the basis of fundamental weights $\varpi_1, \dots, \varpi_4$. They are $(p-4, 1, 1, p-4)$ and $(2, p-2, p-2, 2)$, respectively. Under the stated characteristic assumptions, the first cohomology dimension for $H = \mathrm{PSL}_5(\mathbb{F}_p)$ is read from the normalized standard expansion of the corrected web:

$$\dim_k H^1\big(H, M_{1;2}(p)\big) = [v]$$




$$- (v + v^{-1}) = [v](v + v^3) = 1.$$

For $M_{2;2}(p)$, the four pictured terms in Section 5 give $4 - 2 - 0 - 0 = 2$. The passage from these web coordinates to cohomology uses Lemma 6B.1 and Lemma 6C.1.

For fixed root systems, first cohomology in defining characteristic is bounded by Cline–Parshall–Scott [CPS09], and first extensions by Parshall–Scott [PS11]. For cross characteristic see [GT11, GT20]; for quantitative bounds in growing rank see [PS14]. The ranks in our construction grow. By contrast, the earlier value $469 > 63 = \dim \mathrm{SL}_8$, independently checked by Scott and Sprowl shortly after the June 2012 AIM workshop, already sufficed for the Wall counterexamples discussed in the introduction [G+12, Lüb20]. Varying the field alone need not increase first cohomology.

## 7. Methods, AI assistance and AI declaration

We explain how we found the construction and how AI entered the picture. The route was rather less tidy than the proof, so this section is written in an anecdotal style.

7A. **Our previous work as a starting point.** One prejudice we have acquired over the years is that once things get bad, they tend to get as bad as they possibly can. We have encountered versions of this phenomenon in rather different settings, from the prevalence of torsion in intersection cohomology [BT26] to the failure of standard link invariants to distinguish large families of links [K+25]. Lübeck's computations [Lüb20], with first cohomology dimensions already running into the millions, certainly encouraged that prejudice here. We found it hard to believe that an absolute bound was waiting somewhere beyond them.

This paper grew out of our earlier work [G+26b], where we used planar $\mathfrak{sl}_3$ webs to construct $\mu$-coefficients growing exponentially with $n$ in the symmetric groups $\mathrm{Aut}(\{1, \dots, n\})$. The construction repeats a small piece of web and controls the correction needed to obtain a canonical basis element. A complicated family of permutations thus becomes a fairly simple family of pictures. We wondered whether the same way of thinking could settle Guralnick's conjecture. There was an extra difficulty: large extension dimensions between arbitrary simple modules do not automatically give large first cohomology. For that, one of the modules must be trivial.

So we boldly asked GPT Astra 6.0: "Use our $\mu$-paper to solve Guralnick's conjecture." It first searched the literature and came back with the familiar obstruction: first cohomology requires the lower endpoint to be $w_0$, so an arbitrary family of large $\mu$-coefficients is not enough. It pointed us to [SX10, Section 4, final remark], where Scott–Xi explicitly leave open whether the coefficients $\mu(w_0, w)$ corresponding to first cohomology can be unbounded. Quite right, but hardly news to us. The silly part was treating this as a reason to stop.

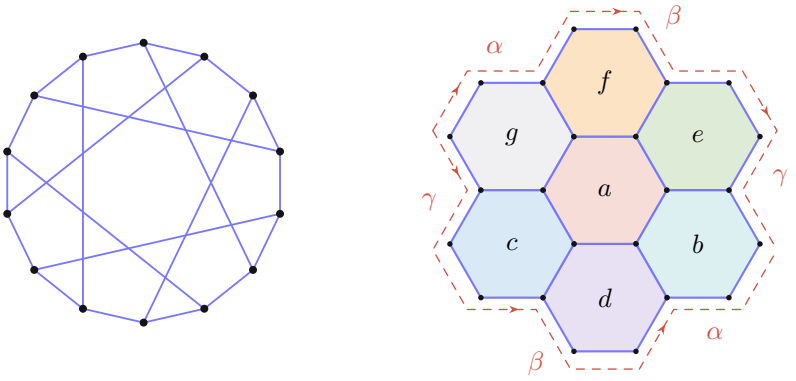
So we took a stick and gave it a poke: "Come on, you do not have to take the request literally. Use the ideas and change the construction if necessary." It then obliged. Unfortunately, we did not understand the answer, and we went back to thinking about what a solution ought to look like.

7B. **Put hexagons on an annulus.** The guiding idea was simple enough: affine = annular. Putting diagrams on an annulus to obtain affine versions of familiar algebras is an old theme, going back several decades; see [HT27, Section 1B] for some historical background. Since the coefficients relevant to first cohomology live in affine KL theory, we should try putting our webs on an annulus.

There was also a more specific picture in mind. The connection between tensor canonical bases and KL polynomials [FKK98] makes the difference between a web and a canonical basis element particularly relevant. KK [KK99] showed that the $\mathfrak{sl}_3$ web basis need not be dual canonical. Their

"hexagon-in-hexagons" picture (a central hexagon surrounded by six others, see the right of Figure 6) is a memorable example of where the two bases start to disagree. Related hexagonal patterns supplied the growing webs in [G+26b]. Hexagons seemed a good place to look for trouble.

When we tried wrapping these pictures around an annulus, they reminded us of the classical embedding of the Heawood graph on a torus. This graph has 14 vertices and 21 edges, and the embedding has seven hexagonal faces. It can itself be drawn as one hexagon surrounded by six others, with suitable boundary identifications; see Figure 6.




The Heawood graph A torus embedding

FIGURE 6. The Heawood graph, drawn in the plane and on a torus. On the right, identify opposite three-edge boundary paths with matching letters and arrows. The dashed red curves mark these pairings. The quotient has 14 vertices, 21 edges and seven hexagonal faces.

Cutting the torus open along a suitable noncontractible curve gives an annulus, with the cut edges ending on its boundary. The Heawood graph therefore looked like a rather tempting starting point: a tiny graph built from exactly the sort of hexagons we were looking for. We tried to turn this picture into a family of annular webs and work out the relevant $\mu$-coefficients. It did not quite work, but the picture still looked promising.

7C. **A little help with the graph.** In the earlier planar calculation, AI assistance had helped us modify an initial web family and find a canonical correction that worked uniformly [G+26b, Section 9]. That suggested trying the same division of labor here. We gave GPT Astra 6.0 the picture and asked it to make the idea work: massage the graph a little if necessary, and try to find a counterexample to Guralnick's conjecture.

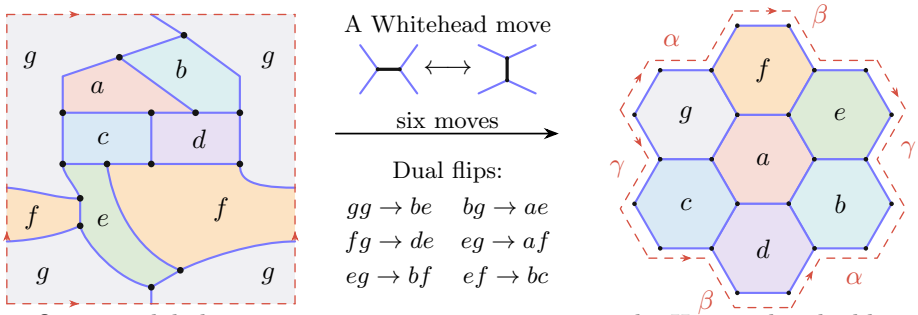
It succeeded fairly quickly. The construction seems to generalize one of Scott–Xi's examples [SX10], and looks rather natural when drawn on an annulus. This time we could understand the proof: the picture helped us see what the formulas were doing. The writing, however, was terrible. We took the argument as a blueprint and worked it into the present paper.

The resulting graph is not the Heawood graph, although its toroidal closure still has 14 vertices and 21 edges; compare Figure 1. The particularly pleasant surprise is that the graph itself need not grow. In the planar construction we repeated web pieces; here we keep the graph and change only the edge thicknesses. This is enough to produce unbounded first cohomology.

*Remark* 7C.1. The resemblance to the Heawood graph is rather concrete: our toroidal closure is just six Whitehead H=I moves away from its standard torus embedding; see Figure 7. ◇

*Remark* 7C.2. Our construction varies only the edge thicknesses. We suspect that things get considerably worse if the web graph itself is also allowed to grow, but we cannot yet prove a stronger growth statement this way. In the large-characteristic setting considered here, Lübeck's argument bounds first cohomology dimensions for restricted irreducible modules by $2^{O(r^3)}$, where $r$ is the rank [Lüb20, Remark (d) after Theorem 4.7]. There is plenty of room between this upper bound and our linear lower bound. ◇

The geometric idea helped us ask a more useful question, and the machine helped turn that idea into a family and a calculation that worked. Responsibility for the mathematics remains ours.




Our toroidal closure
$4, 4, 5, 5, 6, 6, 12$

The Heawood embedding
$6, 6, 6, 6, 6, 6, 6$

FIGURE 7. Six Whitehead moves from our closure to the Heawood embedding. Colors and letters track the seven faces; $pq \to rs$ denotes the edge flip, with the sequence read row by row. Identify opposite dashed sides on the left, and boundary paths with matching letters on the right, following the arrows. The numbers record face lengths.

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